ALGEBRAIC GEOMETRY

github.com/danimalabares/stack

Contents

1. Algebraic sets

The *affine space* $\mathbb{A}^n$ is the Cartesian product of the field n times. An *algebraic set* is the zero set of an ideal in $k[x_1, \dots, x_n]$.

$V(I)$ and lots of properties...

Lemma 1.1. *An algebraic set is irreducible if and only if its defining ideal is prime.*

Proof. First notice that in the case that Y is an irreducible subvariety, this ideal is prime: if $fg \in I_Y(U)$, then $V(f) \cup V(g) \supset Y \cap U$ so that $(V(f) \cup V(g)) \cap Y = Y \cap U$, and then it must be true that $V(f)\cap Y \subseteq V(g)\cap Y$ or that $V(f)\cap Y \supseteq V(g)\cap Y\cap U$ since Y is irreducible, and thus, say, $V(f) \supseteq Y \cap U$, that is, $I(V(f)) \subseteq Y \cap U$ so that $f \in I(Y \cap U)$. $\square$

Theorem 1.2 (Weak Nullstellensatz)**.** *(Misha's formulation.) Let $A \subset \mathbb{C}^n$ be an affine variety and $\mathcal{O}_A$ the ring of polynomial functions on A. Then every maximal ideal in A is an ideal of a point $a \in A$.*

The idea behind the radical idea is that we can take roots of elements:

Definition 1.3. $I \subset R$ is *radical* if for every $x \in R$ we have $x^n \in I \implies x \in I$.

Definition 1.4. An element $a \in R$ is called *nilpotent* if $a^n = 0$.

Exercise 1.5. A/I has no nilpotents if and only if I is a radical ideal.

We may understand the Nullstellensatz as saying that algebraic subsets of an affine variety are the radical ideals of its coordinate algebra.

Theorem 1.6 (Nullstellensatz)**.**

$$I(V(I)) = \sqrt{I}$$

where I is an ideal of $k[x_1, \ldots, x_n]$.

Proof. Prove the Weak Nullstellensatz first and then use Rabinowitsch trick (which uses localization). $\square$

(Prove this:) $k[x_1, \ldots, x_n]/I$ has no nilpotents if and only if $I = \sqrt{I}$.

Now we discuss how to decompose a variety in irreducible components.

Definition 1.7. An algebraic set is *irreducible* if it cannot be decomposed as a union of two properly contained algebraic sets.

Lemma 1.8. *An affine variety is irreducibly if and only if its coordinate ring has no zerodivisors.*

Proof. A zero divisor gives an expression $fg = 0$. $\square$

Exercise 1.9. If $\varphi : A \to B$ is a surjective morphism of varieties and B is irreducible, so is A.

To prove the decomposition in irreducible ideals we must discuss Noetherianity.

Definition 1.10. A ring R is *noetherian* if any ascending chain of ideal stabilizes.

Theorem 1.11 (Hilbert's basis theorem)**.** *If R is a Noetherian ring, so is $R[x]$. Equivalently, every ideal in $R[x]$ is finitely generated.*

Lemma 1.12. *A ring R is Noetherian if and only if all its ideals are finitely generated.*

Proof. Try! $\square$

Theorem 1.13 (Hilbert's basis theorem, Misha's version)**.** *Any finitely generated ring is Noetherian.*

The idea is to use this theorem to show that any algebraic set may be decomposed as a finite union of irreducible varieties. The idea of the proof is to reduce the original algebraic set to a union of two proper subvarieties, and repeat the process until we arrive at irreducible varieties.

Definition 1.14. A topological space is *Noetherian* if any descendending sequence of closed subsets stabilizes.

Definition 1.15. *Zariski topology* is the topology whose closed sets are algebraic sets.

Exercise 1.16 (Misha)**.** Any variety is Zariski compact.

To prove that any variety has a decomposition as a finite union of irreducible components, we first show pointwise that every point is contained in a finite union of irreducible components. With existence at hand we just prove in a similar way the existence of the decomposition for all the variety.

I don't know where to put this yet:

Definition 1.17. The *degree* of an affine or projective variety X of dimension n is the number of points of intersection of X and n hyperplanes in general position.

Exercise 1.18. A projective variety $Y \subseteq \mathbb{P}^n$ has dimension $n-1$ if and only if it is the zero set of a single irreducible homogeneous polynomial f of positive degree.

Proof. Y has dimension $n-1$ if and only if there is a chain of irreducible subsets $Y \supset Y_2 \supset \ldots \supset Y_{n-1}$ if and only if there is a chain of prime ideals $I(Y) \subset I_2 \subset \ldots \subset I_{n-1}$. But $S := k[x_0, \ldots, x_n]$ has Krull dimension $n+1$, so that $I(Y)$ has height 1 (not considering the irrelevant ideal) and thus principal since S is a UFD. $\square$

Proof. ($\implies$) Suppose Y ir irreducible and $\dim Y = n-1$. Pick any nonzero homogeneous $g \in I(Y)$. Factor g into homogeneous irreducible facts and choose one such facto f. Then f is homogeneous, irreducible and vanishes on Y, so $Y \subseteq V(f)$. Both Y and $V(f)$ are irreducible projective varieties and have the same dimension $n-1$, hence $Y = V(I(f))$.

($\impliedby$) Suppose $Y = V(f)$ for some nonconstant homogeneous polynomial f. Then it has dimension $n-1$. $\square$

2. Regular functions, rational functions and morphisms

We start with the intuitive definition, yet different from Harshorne's, which works well for affine varieties. A *regular* or *polynomial map* is $\varphi : V \subset \mathbb{A}^n \to W \subset \mathbb{A}^m$ if there exist $P_1, \ldots, P_m \in k[x_1, \ldots, x_n]$ such that $\varphi(a_1, \ldots, a_n) = (P_1(a_1, \ldots, a_n), \ldots, P_m(a_1, \ldots, a_n))$ and $\varphi(V) \subset W$. If φ is polynomial then there is an induced map in coordinate rings $\varphi^* : k[W] \to k[V]$ given by pullback. This map is well defined since $\varphi^*(I(W)) \subset I(V)$.

A polynomial map $\varphi : V \to W$ is an *isomorphism* if it has a left and right inverse that is also a polynomial function. $V \cong W \iff k[V] \cong k[W]$ for affine varieties.

Here is Hartshorne's definition:

Definition 2.1. Let Y be a quasi-affine variety (i.e. an open set of an affine variety). A function $Y \to k$ is called *regular* at $P \in Y$ if there is an open set $U \ni P$ and $g, h \in k[x_1, \ldots, x_n]$ such that $f = g/h$ on U. f is *regular* if it is regular at all points of Y.

And the definition for projective varieties:

Definition 2.2. Let $Y \subset \mathbb{P}^N$ be a (quasi)projective variety. A function $f : Y \to k$ is regular at $P \in Y$ if there is an open set $U \ni P$ such that $f = g/h$ for two homogeneous polynomials of the same degree, with h nonzero in U.

Note that although homogeneous polynomials are not functions on $\mathbb{P}^n$ (because we can rescale points, but the functions will give different outputs), the quotient of two homogeneous polynomials is (because the rescaling factor cancels out).

Definition 2.3. A *morphism* of varieties $\varphi : V \to W$ is a function that pulls back polynomial functions $f : W \to k$ to polynomial functions $\varphi^* f : V \to k$.

Exercise 2.4. φ is a morphism of affine varieties if and only if it is a polynomial function.

Proof. If φ is polynomial then it is a morphism: if $f : W \to k$ is polynomial, then $\varphi^* f(a_1, \ldots, a_n) = f(P_1(a_1, \ldots, a_n), \ldots, P_m(a_1, \ldots, a_n))$ is also polynomial. Conversely, if φ is a morphism we see that its coordinate functions are polynomials by computing $y_i \varphi^*$ where $y_i : W \to k$ is i-th coordinate function (which is clearly a polynomial map). □

Now we introduce the notion of local ring (stalk). Here's Olivier's (intuitive) definition: Let V be an affine variety (:= irreducible algebraic set) and $p \in V$.

$$\mathcal{O}_p(V) := \{f \in k(V) : \exists g, h \in k[V] \text{ s.t. } f = g/h \text{ and } h(p) \neq 0\} \subset k(V).$$

Compare with Hartshorne's:

Definition 2.5. Let Y be a variety (affine or projective) and $P \in Y$. The *local ring* of P on Y is the ring of germs of regular functions on Y near P.

Exercise 2.6. The local ring is local with maximal ideal given by functions vanishing at P.

Exercise 2.7. Show that if V is an affine variety, $\mathcal{O}_p(V) = k[V]_{\mathfrak{m}_p}$ where $\mathfrak{m}_p$ is the set of regular functions that vanish in p.

Proof. Consider the obvious map: take a quotient of classes $[g]/[h]$ in the localization of the coordinate ring of V and map it to the rational function g/h. This is well defined because if $g - g' \in I(V)$, then they obviously coincide in any open set of V because they coincide in all of V, hence forming the same germ. The key fact is that $\mathcal{O}_p(V)$ is the germs of rational functions of V. It is injective because if a germ g/h is zero, then there's an open of V where g vanishes, implying that $g = 0$ since it is a polynomial (or holomorphic) map. It is obviously surjective. □

Definition 2.8. Let $S = \bigoplus_{d \geq 0} S_d$ be a graded ring and let $f \in S$ be homogeneous. The localization S_f is again graded by

$$\deg(a/f^m) = \deg(a) - m \deg(f)$$

for homogeneous $a \in S$. We write $(S_f)_0$ for its degree-zero part. Thus $(S_f)_0$ consists of the fractions a/f^m such that

$$\deg(a) = m \deg(f).$$

More generally, if $\mathfrak{p}$ is a homogeneous prime of S, we write $S_{(\mathfrak{p})}$ for the degree-zero part of the localization of S at the homogeneous elements outside $\mathfrak{p}$.

Proposition 2.9. *Let $Y = \operatorname{Proj} S$ and let $f \in S$ be homogeneous. Then the open set*

$$D_+(f) = \{\mathfrak{p} \in Y : f \notin \mathfrak{p}\}$$

is affine, with coordinate algebra

$$\mathcal{O}_Y(D_+(f)) = (S_f)_0.$$

In other words, restricting to the chart where f does not vanish corresponds to localizing at f and taking degree zero.

Theorem 2.10. *Let $Y \subseteq \mathbb{P}^n$ be a projective variety with homogeneous coordinate ring $S(Y)$. Then:*

(1) $\mathcal{O}(Y) = k$,
(2) $\mathcal{O}_P = S(Y)_{(\mathfrak{m}_P)}$, *where $\mathfrak{m}_P$ is the ideal generated by the homogeneous $f \in S(Y)$ not vanishing in P,*
(3) $K(Y) \cong S(Y)_{((0))}$.

Now let's discuss the equivalence of categories between affine varieties and finitely generated reduced algebras.

Proposition 2.11. *The coordinate algebra of an affine variety V, i.e. $k[x_1, \ldots, x_n]/I(V)$ is isomorphic to the ring of regular functions on V, i.e. $\mathcal{O}(V)$.*

The following proposition is also at 01I1

Proposition 2.12. *Let $V, W \subset \mathbb{A}^n$ are irreducible algebraic set. There is a bijection*

$$\operatorname{Hom}_{k-alg}(k[W], k[V])) \xrightarrow{bij} \underbrace{\operatorname{Mor}(V, W)}_{polynomial\ maps}.$$

Proof. We map

$$\begin{aligned} \tilde{\varphi} : \underbrace{k[W]}_{=k[y_1,\ldots,y_m]/I(W)} &\longrightarrow \underbrace{k[V]}_{=k[x_1,\ldots,x_n]/I(V)} \\ y_1 &\longmapsto f_1 \\ &\vdots \\ y_m &\longmapsto f_m \end{aligned}$$

to

$$\begin{aligned} \varphi : V &\longrightarrow W \\ (x_1, \ldots, x_n) &\longmapsto (f_1(x_1, \ldots, x_n), \ldots, f_m(x_1, \ldots, x_n)). \end{aligned}$$

We check that this is well-defined, i.e. that it doesn't depend on the choice of representatives, namely the f_i. But choosing other f_i which differ by an element of the ideal would not make a difference when evaluating at V because the ideal is precisely $I(V)$.

The inverse map we already described above. □

This shows that we have an equivalence of categories:

Lemma 2.13. *There is an equivalence of categories between finitely generated reduced k-algebras and irreducible algebraic sets.*

Now let's discuss rational functions. Again let's start with a more reasonable explanation. Since $k[V]$ for an irreducible affine algebraic set is a domain, there exists a fraction field, which is the set of *rational functions*. Functions here look like f/g for $f, g \in k[V]$, which are not defined in $V(g)$. Since this can lead to different ways of defining the same function — the correct definition shall be that we take the greatest set where the function can be defined.

Definition 2.14. The *ring of rational functions* is the set of regular functions defined on open sets with the identifications that $(U, f) \sim (V, g)$ if they coincide in an intersection. (Like the stalk, but there's no point.)

Compare this to Voisin's way of describing a meromorphic function: it's a function that can be written locally as a quotient of two holomorphic functions.

3. Dimension: Krull dimension, transcendence degree

Definition 3.1. A nonempty subset Y of a topological space X is *irreducible* if it cannot be expressed as a union of two proper subsets, each of which is closed in Y.

Definition 3.2. If X is a topological space, the *dimension* of X is the supremum of all integers n such that there exists a chain $Z_0 \subset Z_1 \subset \ldots \subset Z_n$ of distinct irreducible closed subsets of X. The *dimension of an algebraic variety* is its dimension as a topological space.

Definition 3.3. In a ring A, the *height* of a prime ideal $\mathfrak{p}$ is the supremum of all integers n such that there exists a chain $\mathfrak{p}_0 \subset \mathfrak{p}_1 \subset \ldots \subset \mathfrak{p}_n = \mathfrak{p}$ of distinct prime ideals. The *Krull dimension* of A is the supremum of the heights of all prime ideals.

Proposition 3.4. *If Y is an affine algebraic set, the dimension of Y is equal to the dimension of its coordinate ring.*

Theorem 3.5. *Let k be a field and let B be an integral domain which is a finitely generated k-algebra. Then:*

(1) *the dimension of B is equal to the transcendence degree of the quotient field $K(B)$ of B over k;*
(2) *For any prime ideal $\mathfrak{p}$ in B, we have*

$$\textit{height}\ \mathfrak{p} + \dim B/\mathfrak{p} = \dim B.$$

Exercise 3.6. Let $Y \subseteq \mathbb{A}^3$ be the set $Y = \{(t, t^2, t^3 : t \in k\}$. Show that Y is an affine variety of dimension 1. Find generators for the ideal $I(Y)$. Show that $A(Y)$ is isomorphic to a polynomial ring in one variable over k. We say that Y is given by the *parametric representation* $x = t$, $y = t^2$, $z = t^3$.

Proof. It's immediate that $x^2 - y$ and $x^3 - z$ must be in the ideal defining Y. Let $J = (x^2 - y, x^3 - z)$. Consider the ring map

$$\begin{aligned} \varphi : k[x, y, z] &\longrightarrow k[t] \\ x &\longmapsto t \\ y &\longmapsto t^2 \\ z &\longmapsto t^3 \end{aligned}$$

Since φ is given literally by the parametrization of Y, it follows from definition that the ideal of Y is the kernel of φ.

It's clear the in the kernel of φ lie the polynomials $x^2 - y$ and $x^3 - z$. But how to make sure that J is the complete kernel?

The procedure by AI is the following. First notice that there is an isomorphism $k[x, y, z]/J \cong k[x]$ given by $f(x, y, z)+J \mapsto f(x, x^2, x^3)$. This shows that $k[x, y, z]/J$ is a domain, and thus J is prime. Further, the projection $k[x, y, z] \to k[x, y, z]/J$, which obviously has kernel J, is exactly the map I introduced before from $k[x, y, z]$ to $k[t]$ when we see $k[x, y, z]/J$ as $k[t]$. Since the kernel of such map is the ideal of Y by definition, we are done. □

4. Zariski tangent space, smooth points

Let's discuss Zariski tangent space. The simplest notion of nonsingularity is what we expect:

Definition 4.1. Let $Y \subseteq \mathbb{A}^n$ be ann affine variety, and let $f_1, \dots, f_t \in R = k[x_1, \dots, x_n]$ be a set of generators for the ideal of Y. Y is *nonsingular at a point* $P \in Y$ if the rank of the matrix $\left(\frac{\partial f_i}{\partial x_j}(P)\right)$ is $n - r$, where r is the dimension of Y. Y is *nonsingular* if it is nonsingular at every point.

It can be checked that this definition is independent of the choice of generators. But we look for a criterion that is more algebraic.

Definition 4.2. Let R be a ring and $\mathfrak{m}$ a maximal ideal of R. The *residue field* is $R/\mathfrak{m}$.

Definition 4.3. Let R be a noetherian local ring with maximal ideal $\mathfrak{m}$ and residue field $k := R/\mathfrak{m}$. R is a *regular local ring* if $\dim_k \mathfrak{m}/\mathfrak{m}^2 = \dim R$.

Upshot. $\mathfrak{m}$ is the ideal of functions vanishing at P. This means that the constant term of $h \in \mathfrak{m}$ is zero. Differentiate h. Now the constant term is what originally was the linear part. If h' has no constant term, that is, if h vanishes at P up to order 1, then h can be seen as a product $h = fg$ for $f, g \in \mathfrak{m}$. Indeed, the smallest nonzero term of h is of degree 2, so we can make it look like a product of two things of degree 1. Conversely, products $h = fg$ vanish up to order 1 since the first-order coefficient is given by the derivative (Taylor expansion) and $dh = df \cdot g + f dg = 0$ at P.

Then we can interpret $\mathfrak{m}/\mathfrak{m}^2$ as follows. Among all the functions which vanish at P, i.e. inside $\mathfrak{m}$, we identify any two functions if their difference is in $\mathfrak{m}^2$, that is, if their difference vanishes up to order 1 at P, that is, if their linear parts coincide at P.

Thus, $\mathfrak{m}/\mathfrak{m}^2$ is the space of linear parts of polynomials vanishing at P.

Theorem 4.4. *Let $Y \subseteq \mathbb{A}^n$ be an affine variety. Let $P \in Y$ be a point. Then Y is nonsingular at P if and only if the local ring $\mathcal{O}_{P,Y}$ is a regular local ring.*

Proof. Still hard to grasp completely, but an important insight is that the map

$$\begin{aligned} \nabla : k[x_1, \dots, x_n] &\longrightarrow k^n \\ f &\longmapsto \nabla f = \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n}\right) \end{aligned}$$

has kernel $\mathfrak{m}^2$ and is surjective when restricted to $\mathfrak{m}$ (because $\nabla(x_i - p_i) = (0, \dots, 0, 1, 0, \dots 0)$, so we construct a base of k^n) so we have the isomorphism

$$\mathfrak{m}/\mathfrak{m}^2 = k^n,$$

which we understand as the space of linear directions at P. □

Definition 4.5. Let Y be any variety, Y is *nonsingular* at a point $P \in Y$ if the local ring $\mathcal{O}_{P,Y}$ is a regular local ring, that is, if $\dim_k \mathfrak{m}/\mathfrak{m}^2 = \dim \mathcal{O}_{P,Y}$.

Now we discuss singularity theory for plane curves.

Upshot: the multiplicity of a point is how many tangent lines there are at the point.

Smooth points of a plane curve are those with multiplicity 1. The multiplicity of a plane curve F at the origin is the degree of the lowest nonzero homogeneous term F_m of $F = F_m + F_{m+1} + \ldots + F_n$. The tangent lines at the origin are the linear factors of F_m, which decomposes in linear components by the following homogenization trick:

Exercise 4.6. Let f_m be a homogeneous polynomial of degree m. Show that f_m is a product of m linear factors $f_m = \prod(b_i x + c_i y)$.

Proof. Upshot: fix $y = 1$, factor as product of linear coefficients using fundamental theorem of algebra, and then rehomogenize. The last is because a 1-variable polynomial F is homogeneous of degree m if and only if $F(\lambda x) = \lambda^m F(x)$. So if you have $F(x) = f(x, 1)$ you rehomogenize by putting $y^m F(x/y) = y^m f(x/y, 1)$. Which is strange because $y^m F(x/y)$ is in $k[x, y, 1/y]$, but that doesn't matter because when you plug it in you get

$$y^m f(x/y, 1) = y^m c \prod(\frac{x}{y} - \alpha_i) = c \prod(x - \alpha_i y),$$

(where c is a constant coming from making the product having no coefficients on x after using fundamental theorem of algebra) which is already a product of linear polynomials. We conclude by showing that in fact $y^m f(x/y, 1) = f(x, y)$, but that's just literally

$$y^m f(x/y, 1) = y^m \sum_i a_i \frac{x^i}{y^i} = \sum_i a_i x^i y^{m-i}.$$

□

Thus tangent lines can have multiplicity. A *node* is an ordinary double point, which means that there are two tangent lines and they are not the same. A *cusp* is a double point with a double tangent line.

Exercise 4.7 (Intersection multiplicity)**.** If $Y, Z \subseteq \mathbb{A}^2$ are two distinct curves, given by equations f and g, and if $P \in Y \cap Z$, we define the *intersection multiplicity* $(Y \cdot Z)_P$ of Y and Z at P to be the length of the $\mathcal{O}_P$-module $\mathcal{O}_P/(f, g)$.

- Show that $(Y \cdot Z)_P$ is finite, and $(Y \cdot Z)_P \geq \mu_P(Y)\mu_P(Z)$.

The upshot about Bézout theorem is that the "size" of the intersection of two curves is the product of their degrees. In nice cases this is just the number of intersection points. In even nicer cases one of the curves is a line, in which case the product of degrees is just the degree of the curve, so that we compute the degree of a curve by *the number of intersection points with a hyperplane.*

But the notion of "size" of the intersection has to be made precise, which is done via the notion of multiplicity. The idea is that intersecting with a line $\ell = ax+by+c$ where, say, $b \neq 0$, can be done by substituting y for $-\alpha x - b$. This makes the "resultant" $f_\ell(x) := f(x, \alpha x + \beta)$ be a polynomial in one variable, which can be factored as a product of linear terms, each of the roots of which are the intersection points, and are now equipped with a multiplicity.

Definition 4.8. The *multiplicity* or *intersection index* of $f \in k[x,y] \setminus k$ and the line $\ell = ax + by + c$ (where $a \neq 0$ or $b \neq 0$) is

$$(\ell, f)_p = \begin{cases} 0 & \text{if } p \notin V(f, \ell) \\ \infty & \text{if } p \in V(\ell) \text{ and } V(\ell) \subset V(f) \\ m_i & \text{if } p = (x_i, \alpha x_i + \beta) \end{cases}$$

where $f_\ell(x) := f(x, \alpha x + \beta)$ factors as $c \prod (x - x_i)^{m_i}$.

If $V(\ell) \not\subset V(f)$, the integer

$$\deg(f) - \sum_{i=1}^{r} m_i$$

is the *multiplicity of* ℓ, f *at infinity.*

Exercise 4.9. If $f = \sum_{i=0}^{d} f_i \in k[x,y]$ for f_i homogeneous of degree i, an *asymptotic direction* of f is $ax + by$ which divides f_d.

Show that $m_\infty \neq 0$ if and only if $y - \alpha x$ is an asymptotic direction of f.

Proof. This uses a theorem using the resultant... $\square$

Theorem 4.10 (Bézout)**.** *Let k be an algebraically closed field. Let $C, P \subset \mathbb{P}^2_k$ be two curves without irreducible components in common and suppose that $C \cap D = \{p_1, \ldots, p_r\}$ with intersection multiplicities $m_1, \ldots, m_r$ in $p_1, \ldots, p_r$. Then*

$$degf\, degg = \sum_{i=1}^{r} m_i.$$

Proposition 4.11. *Let $f \in k[x,y] \setminus k$ and $p \in V(f)$. There exists an integer $m_p(f) \geq 1$ called the multiplicity of p in f such that for every line ℓ passing through p*

$$(\ell, f)_p \geq m$$

and $(\ell, f)_p > m$ for at most m lines and at least one line.

Definition 4.12. Let $p = (x_0, y_0) \in V(f)$. Write $f(x + x_0, y + y_0) = f_m(x,y) +$ lower degree terms, i.e. translate f so that the origin is in its zero set and write as homogeneous polynomials. Then put $f_m(x,y) = \prod_{i=1}^{r} (a_i x + b_i y)^{m_i}$. The lines $\ell_i = a_i x + b_i y$ are called *tangent lines* to f and each has a multiplicity m_i.

Definition 4.13. A point $p \in V(f)$ is *smooth* if $m_p(f) = 1$ and *ordinary* if all its tangent lines have multiplicity 1.

Thus, ordinary points are singular, but these are mild singularities. Essentially they are points where m straight lines intersect transversally.

Definition of intersection index of two plane curves without common components. Definition of intersection index of two plane curves with common components. Properties of intersection index (7 of them). These 7 properties characterize uniquely the intersection index.

The multiplicity of a root α of a power series $f \in k[[x]]$, i.e. the greatest integer such that $(x - \alpha)^r | f$, equals $\dim_k k[[x]]/(x^r) = r$. (Here's why.) This leads to the alternative definition of intersection index as $\dim_k k[[x,y]]/(f(x - x_0, y - y_0), g(x - x_0, y - y_0))$. This satisfies the 7 properties of the intersection index.

Proposition 4.14 (Jacobian criterion)**.** *The singular locus of $V(f)$ is $V(f, \partial_x, \partial_y)$.*

Theorem 4.15 (Plücker formulas)**.** *Let $C \subset \mathbb{P}^2$ be an irreducible curve of degree d whose only singularities are δ nodes and χ ordinary cusps (cusps such that the tangent line intersects with multiplicity 3 at the singular point). Let*

- *i be the number of inflection lines (lines ℓ such that $(\ell, C)_p \geq 3$, and, in this case, do not intersect another point with multiplicity ≥ 2), and*
- *d^* the number or lines tangent to C passing through a point $p \notin C$ such that p is not in any tangent lines at the singular points, tangent inflection lines or bitangent lines (lines that are tangent to two different points).*

Then

$$\begin{aligned} d(d-1) &= d^* + 2\delta + 3\chi \\ 3d(d-2) &= i + 6\delta + 8\chi. \end{aligned}$$

5. Rational maps and blow up

The category of varieties and dominant rational maps is equivalent to the category of finitely generated field extensions of k with the arrows reversed.

Exercise 5.1. Let Y be the cuspidal curve $y^2 = x^3$ in $\mathbb{A}^2$. Blow up the point $O = (0,0)$, let E be the exceptional curve, and let $\tilde{Y}$ be the strict transform of Y. Show that E meets $\tilde{Y}$ in one point, and that $\tilde{Y} \cong \mathbb{A}^1$.

6. Normalization

All bounded meromorphic functions are holomorphic.

Definition 6.1. Let R be a domain. An element of the fraction field $\mathrm{Frac}(R)$ is *integral* if it is the root of a monic polyomial with coefficients in R.

R is *integrally closed* if it coincides with its integral closure, that is, the set of elements of $\mathrm{Frac}(R)$ which are integrally closed over R.

Definition 6.2. A variety is called *normal at P* is $\mathcal{O}_P$ is integrally closed. It is called *normal* if every point is normal.

Exercise 6.3. If Y is an affine variety, $\mathcal{O}_P$ is intergrally closed for all $P \in Y$ if and only if $\mathcal{O}_Y$ is integrally closed.

Proof. For the converse let $f/g \in \mathrm{Frac}\mathcal{O}_P$ be integral. Then there exists $P \in \mathcal{O}_P[t]$ monic killing f/g. We want to show that $f/g \in \mathcal{O}_P$ for which it's enough to put f/g in $\mathcal{O}_Y$ using the closure-integrality of Y. Suppose

$$P = \sum \frac{f_i}{g_i} t^i.$$

Since this polynomial is not in $\mathcal{O}_Y[t]$, we cannot use it directly to prove integrality of f/g; in fact f/g might not be integral at all. So we get rid of the fractions considering the least common multiple of the g_i. But if we just multiply by the lcm we get a polynomial that is not monic. We must consider

$$P \cdot lcm^{\deg P}\left(\frac{f}{g}\right) = \sum lcm^{\deg P} \frac{f_i}{g_i}\left(\frac{f}{g}\right)^i = \frac{f_i}{g_i}\left(lcm\frac{f}{g}\right)^{\deg P} + \frac{f_i}{g_i} lcm \left(lcm\frac{f}{g}\right)^{\deg P - 1} + \ldots + \frac{f_0}{g_0} lcm^{\deg P},$$

which is monic over $\mathcal{O}_Y$ and kills $lcm\frac{f}{g}$, which implies that $\frac{f}{g} \in \mathcal{O}_Y$ and thus g is never zero (or simply a unit of the coordinate algebra), and thus f/g is in $\mathcal{O}_P$.

The direct implication is actually easier. Let $f/g \in \mathrm{Frac}\mathcal{O}_Y$ be integral, i.e. there is a polynomial $P \in \mathcal{O}_Y[t]$ killing f/g. Pick any point $p \in Y$. Then P can be taken to be a polynomial in $\mathcal{O}_p[t]$, which in fact kills the fraction $\overline{f/g} \in \mathrm{Frac}\mathcal{O}_p$ in the local ring of p. Since $\mathcal{O}_p$ is integrally closed, $f/g \in \mathcal{O}_P$. Thus, thinking of f/g as a meromorphic function, we see that $g(p) \neq 0$ for all p, meaning the fraction f/g is in fact regular. Alternatively, thinking of f/g as an element of the localization by the maximal ideal $\mathfrak{m}_P$, saying that $f/g \in \mathcal{O}_P$ means that $g \notin \mathfrak{m}_P$ for all P, which is to say that g is in no maximal ideal of $k[Y]$, which means that g is a unit. □

Exercise 6.4.
(1) Show that every conic in $\mathbb{P}^2$ is normal.
(2) Show that the quadric surfaces Q_1, Q_2 in $\mathbb{P}^3$ given by $Q_1 : xy - zw$ and $Q_2 : xy = z^2$ are normal.
(3) Show that the cuspidal cubic $y^2 = x^3$ in $\mathbb{A}^2$ is not normal.

Proof.
(1)
(2)
(3) The element x^2/y has a minimal polynomial in the local ring at the origin, namely $P(t) = t - x$, but it clearly does not lie in the local ring since y vanishes at $(0,0)$.

□

Exercise 6.5. Let Y be an affine variety. Show that there is a normal affine variety $\tilde{Y}$, and a morphism $\pi : \tilde{Y} \to Y$, with the property that whenever Z is a (affine) normal variety and $\varphi : Z \to Y$ is a dominant morphism, then there is a unique morphism $\theta : Z \to \tilde{Y}$ such that $\varphi = \pi \circ \theta$.

Proof. Let $\tilde{Y} = \operatorname{Spec} \tilde{A}$ where $\tilde{A}$ is the integral closure of $A = \mathcal{O}_Y$. Then by Exercise 6.3 it is normal. Now if the dominant map $\varphi : Z \to Y$ induces an injective algebra morphism $\varphi^{\#} : A \to B$ and B is integrally closed, we can put $\theta^{\#} : \tilde{A} \to B$ by the inclusion. Since any element $x \in \tilde{A}$ is the root of a unique monic polynomial $P \in A[t] \subset B[t]$, and since $\theta^{\#}(x)$ must be the root of a monic polynomial in $B[t]$ because B is integrally closed, we conclude that $\theta^{\#}(x)$ is also a root of P. ... but what if the roots are different?

A correct solution is as follows. The inclusion $A \hookrightarrow B$ induces an inclusion on fraction fields $\mathrm{Frac}A \hookrightarrow \mathrm{Frac}B$. The key observation is that by definition an element in the integral closure $\tilde{A}$ of A is an element in the fraction field $\mathrm{Frac}A$. This says that any integral element can already be seen as an element in $\mathrm{Frac}B$. But since B is integrally closed and the minimal polynomial of any such element has coefficients in $A \subset B$, we are done: an integral element of A is mapped to an element of B making the following diagram commute

$$\begin{tikzcd} A \arrow[r, hook, "\varphi^\sharp"] \arrow[dr, "\pi^\sharp"'] & B \\ & \tilde{A} \arrow[u, dashed, "\theta^\sharp"'] \end{tikzcd}$$

For uniqueness, suppose that $\theta' : \tilde{A} \to B$ is another map making the above diagram commute. Then for any $x = a/b$ integral over A we have $\theta'(b)\theta'(x) = \theta'(a)$. But this coincides on A with $\theta^\sharp$ we introduced before, giving $\theta^\sharp(b)(\theta'(x) - \theta^\sharp) = 0$. Since there are no zerodivisors on B, we obtain $\theta \equiv \theta^\sharp$. □

Exercise 6.6. A projective variety $Y \subseteq \mathbb{P}^n$ is *projectively normal* (with respect to the given embedding) if its homogeneous coordinate ring $S(Y)$ is integrally closed.

(1) If Y is projectively normal then Y is normal.
(2) There are normal varieties (i.e. stalks are integrally closed) in projective space which are not projectively normal. For example, let Y be the twisted quartic curve in $\mathbb{P}^3$ given parametrically by $(x, y, z, w) = (t^4, t^3u, tu^3, u^4)$.
(3) Show that the twisted quartic curve Y above is isomorphic to $\mathbb{P}^1$, which is projectively normal. Thus projective normality depends on the embedding.

Proof. (1) Take an affine chart $D_+(f)$. By Proposition 2.9 the coordinate algebra of Y at this open set is given by $(S(Y)_f)_0$, the degree-0 elements in the localization $S(Y)_f$. By Commutative Algebra Proposition **??**, localization of an integrally closed ring remains integrally closed, thus $S(Y)_f$ is integrally closed. Let $A = (S(Y)_f)_0$. If $x \in \mathrm{Frac}(A)$ is integral over A, then it is integral over $S(Y)_f$. Thus $x \in S(Y)_f$. But x is a quotient of degree-0 elements, hence has degree 0. Therefore $x \in (S(Y)_f)_0 = A$.

(2) By the next item, the twisted quartic is isomorphic to $\mathbb{P}^1$, hence is normal. It remains to see that it is not projectively normal in this embedding. Its homogeneous coordinate ring is

$$R = k[t^4, t^3u, tu^3, u^4] \subset k[t, u].$$

The element t^2u^2 lies in $\mathrm{Frac}(R)$, since

$$t^2u^2 = \frac{(t^3u)(u^4)}{tu^3}.$$

Moreover it is integral over R, since

$$(t^2u^2)^2 = (t^3u)(tu^3) \in R.$$

But $t^2u^2 \notin R$: the exponent $(2, 2)$ is not in the semigroup generated by

$$(4, 0), (3, 1), (1, 3), (0, 4).$$

Thus R is not integrally closed.

(3) Let

$$\varphi : \mathbb{P}^1 \to Y$$

be the map

$$[t : u] \longmapsto [t^4 : t^3u : tu^3 : u^4].$$

We construct its inverse. On the open subset of Y where $(x, y) \neq (0, 0)$, define

$$[x : y : z : w] \longmapsto [x : y].$$

On the open subset where $(z, w) \neq (0, 0)$, define

$$[x : y : z : w] \longmapsto [z : w].$$

These opens cover Y. On the image of φ we have

$$[x : y] = [t^4 : t^3u] = [t : u]$$

and

$$[z : w] = [tu^3 : u^4] = [t : u].$$

Thus the two local formulae agree on overlaps and define a morphism $Y \to \mathbb{P}^1$ inverse to φ. Hence $Y \cong \mathbb{P}^1$. Finally, the standard embedding of $\mathbb{P}^1$ has homogeneous coordinate ring $k[t,u]$, which is integrally closed. Thus $\mathbb{P}^1$ is projectively normal.

□

7. Divisors, invertible sheaves

Definition 7.1. A *Weil divisor* on the variety X is a formal sum of irreducible subvarieties over the integers.

More formally, a *prime divisor* is an integral closed subscheme $Z \subset X$ of codimension 1. A *Weil divisor* is a formal sum $D = \sum n_Z Z$ where the sum is over prime divisors of X and the collection $\{Z | n_Z \neq 0\}$ is locally finite. If all coefficients are non-negative we say the divisor is *effective*.

A reasonable notion of order of vanishing should then be introduced. In the case of Riemann surfaces this can be done using the order of vanishings of the denominator and numerator functions in $f = g/h$ where $f \in K(X)^*$. A more general construction is a bit more involved.

Definition 7.2. Let X be a Noetherian integral scheme. Let $f \in R(X)^*$. The *principal Weil divisor associated to* f is the Weil divisor

$$\mathrm{div}(f) = \mathrm{div}_X(f) = \sum \mathrm{ord}_Z(f)[Z]$$

where the sum is over prime divisors and $\mathrm{ord}_Z(f)$ is as in Definition ?. This makes sense by Lemma ? [a lemma about finiteness of the sum].

Definition 7.3. Let X be a locally Noetherian scheme. The *Weil divisor class group* of X, denoted $\mathrm{Cl}(X)$, is the quotient of the group of Weil divisors by the subgroup of principal Weil divisors.

The next section, 02SE, shows there is an injective homomorphism $\mathrm{Pic}(X) \to \mathrm{Cl}(X)$ which is also surjective if X is normal and the local rings of X are UFDs (which happens for smooth varieties).

Earlier, in 01WQ, we have

Definition 7.4. Let S be a scheme.

- A *localy principal closed subscheme* of S is a closed subscheme whose sheaf of ideals is locally generated by a single element.
- An *effective Cartier divisor* on S is a closed subscheme $D \subset S$ whose ideal sheaf $\mathcal{I}_D \subset \mathcal{O}_S$ is an invertible $\mathcal{O}_S$-module.

Followed by Lemma 01WS, which essentially says that a Cartier divisor is cut locally by a single function. After this we would show the well-known construction of sheaf associated to an effective Cartier divisor and eventually construct all group operations and subtleties to arrive at Lemma 01X0, which says that effective Cartier divisors on a scheme are the same as invertible sheaves with fixed regular global section.

Here's two basic facts I'm not sure how to prove:

(1)
$$\mathcal{O}_{\mathbb{P}^n}(1) \cong \mathcal{O}_{\mathbb{P}^n}(H)$$
where $\mathcal{O}_{\mathbb{P}^n}(1)$ is defined as dual of the tautological bundle, and H is any hyperplane (zeroes of a homogeneous degree-1 polynomial in $n+1$ variables)

seen as an effective Cartier divisor, so that $\mathcal{O}_{\mathbb{P}^n}(H)$ is its associated line bundle.

(2)
$$H^0(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(1)) \cong k[x_0, \dots, x_n]_1,$$
that is, the space of sections of $\mathcal{O}_{\mathbb{P}^n}(1)$ is isomorphic to the space of homogeneous polynomials of degree 1 in $n+1$ variables (each of which is a hyperplane in $\mathbb{P}^n$). I think this might be here.

Exercise 7.5. Show that a hypersurface in $\mathbb{P}^3$ given by the zeroes of a homogeneous polynomial of degree d has degree d, i.e., if $S \subset \mathbb{P}^3$, $S = V(f)$ with f homogeneous of degree d, then
$$\mathcal{O}_S(1).\mathcal{O}_S(1) = d.$$

Proof. The exercise is kind of by definition no? The intersection form counts the size of the intersection when the two classes are just two hypersurfaces intersecting transversally (as is the case of two hyperplane sections in general position, both of which have $\mathcal{O}_S(1)$ as associated line bundle). The intersection has d points because its a line in $\mathbb{P}^3$ intersecting a degree-d polynomial. And that's it. □

Exercise 7.6. Let $S \subset \mathbb{P}^3$ be a surface of degree d and $L \subset S$ a line. Compute L^2 in S. Check that when $d = 3$, $L^2 = -1$!

Proof. We apply the genus formula:
$$(L.L + K_S) = 2p_a - 2$$
By L being a line we know that $p_a = 0$ and that the intersection number of L with any divisor of degree d' will be d'. Indeed, $L.d'H = d'(L.H) = d'$ because $L.H$ is the degree of the line (by definition of degree, namely the intersection number of L with as many hyperplanes as its dimension). Thus
$$\begin{aligned} -2 &= (L.L + K_S) = L^2 + (L.K_S) = L^2 + (L.\mathcal{O}_S(d-4)) = L^2 + d - 4 \\ \iff\ & L^2 = -d + 2. \end{aligned}$$
where the canonical bundle of S is computed using adjunction formula $K_S = (K_{\mathbb{P}^3} + N)|_S = (\mathcal{O}_{\mathbb{P}^3}(-4) + \mathcal{O}_{\mathbb{P}^3}(d))$. □

Exercise 7.7 (Bézout)**.** Show that if C_d and $C_{d'}$ are two plane curves (i.e. contained in $\mathbb{P}^2$) of degrees d and d' then $C_d.C_{d'} = dd'$.

Proof. Just notice that the line bundles associated to C_d and $C_{d'}$ are dH and $d'H$, then use linearity of the bilinear of form. To check why the divisor associated to C_d is indeed dH note that the intersection number of C_d and a line H is d because C_d is given by the roots of a polynomial of degree d, thus $C_d.H = d$, and since the Picard group of $\mathbb{P}^2 = \mathbb{Z}H$ we must have $C_d \sim dH$. □

8. Adjunction formulas

There are several statements called adjunction formula in different texts. All of them concern "subvarieties", that is, closed embedded subschemes.

Exercise 8.1 (Genus formula for a curve on a surface)**.** Let $C \to X$ be a closed embedded subscheme of dimension 1 (as a topological space, i.e. pure dimension) inside a smooth surface X. Then $2p_a - 2 = (\mathcal{O}_X(C), \mathcal{O}_X(C))$.

Proof. Consider the ideal sheaf exact sequence

$$0 \longrightarrow \mathcal{O}_X(-C) \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_C \longrightarrow 0$$

This sequence splits since there is an obvious inverse morphism to the inclusion $\mathcal{O}_X(-C) \to \mathcal{O}_X$, namely mapping a function f to Then $\mathcal{O}_X \cong \mathcal{O}_X(-C) \oplus \mathcal{O}_C$. $\chi(\mathcal{O}_X) = \chi$ □

9. Linear systems

A basic fact is that

$$H^0(\mathcal{O}(d)) \cong \{\text{homogeneous polynomials of degree } d\}.$$

And more generally,

$$|D| = \mathbb{P}H^0(\mathcal{O}(D)).$$

Following [**?**, Section 5.2], let $P_1, \ldots, P_n$ be points in $\mathbb{P}^2$. Define

$$V(d, r_1P_1, \ldots, r_nP_n) := \{\text{curves } F \text{ of degree } d : m_{P_i}F \geq r_i \forall i\}.$$

This space has an interesting property: when we blow up the n points, we obtain

$$H^0(X, \mathcal{O}(dH - \sum r_iE_i)) = \{f \in H^0(\mathbb{P}^2, \mathcal{O}(d)) : m_{P_i}(f) \geq r_i\}.$$

Recall (proof?) that the canonical divisor of the blow up is $K_{Bl} = \pi^*K_{\mathbb{P}^2} + E_1 + \ldots + E_n = -3H + \sum E_i$. And the projectivization of the sections of this canonical divisor are exactly $V(-3, P_1, \ldots, P_n)$, which makes no sense because there are no curves of degree -3, that is the canonical divisor of the blow up has no sections.

Exercise 9.1. Let C_1, C_2 be two cubic curves in $\mathbb{P}^2$. Compute the Hodge numbers of the blowup of the 9 points of intersection.

Proof. The solution is:

$$\begin{array}{ccccc} & & 1 & & \\ & 0 & & 0 & \\ 0 & & 10 & & 0 \\ & 0 & & 0 & \\ & & 1 & & \end{array}$$

Which follows from:

- $h^{0,0} = 1$ because X is connected.
- $h^{1,0} = 0$ because X is simply connected.
- $h^{2,0} = 0$ because the canonical bundle of X has no sections.
- $h^{1,1} = 10$ because in the exponential exact sequence we get $\mathrm{Pic} \cong H^2(X, \mathbb{Z})$ (by vanishing irregularity and genus) and we know $\mathrm{Pic} = \mathbb{Z}H \oplus \mathbb{Z}E_1 \oplus \ldots \oplus \mathbb{Z}E_9$.

□

10. Ample line bundles

First is this lemma that comes from modules.tex. I think these sets X_s are the base points of the bundle. Because look: image of s just means consider the section s of the line bundle as a germ near x. Now a line bundle is a locally free rank-1 $\mathcal{O}_X$-module, so its sections, like s, may be multiplied by germs of functions in the maximal ring $\mathfrak{m}_x$, i.e. the functions that vanish at x. So X_s is the vanishing locus of the section s. If $s(x) \neq 0$, obviously $s \notin \mathfrak{m}_x\mathcal{L}_x$, so $x \in X_s$. Conversely, I would like to show that if $s(x) = 0$ then $s \in \mathfrak{m}_x\mathcal{L}_x$ but I'm not sure how. It's like: a vector field with a zero can be multiplied by a function that vanishes at the point, sure, but what's this function?

Lemma 10.1. *From modules.tex. Let X be a ringed space. Assume that each stalk $\mathcal{O}_{X,x}$ is a local ring with maximal ideal $\mathfrak{m}_x$. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module. For any section $s \in \Gamma(X, \mathcal{L})$ the set*

$$X_s = \{x \in X \mid \textit{image } s \notin \mathfrak{m}_x\mathcal{L}_x\}$$

is open in X. The map $s : \mathcal{O}_{X_s} \to \mathcal{L}|_{X_s}$ is an isomorphism, and there exists a section s' of $\mathcal{L}^{\otimes -1}$ over X_s such that $s'(s|_{X_s}) = 1$.

Proof. Suppose $x \in X_s$. We have an isomorphism

$$\mathcal{L}_x \otimes_{\mathcal{O}_{X,x}} (\mathcal{L}^{\otimes -1})_x \longrightarrow \mathcal{O}_{X,x}$$

by Lemma **??**. Both $\mathcal{L}_x$ and $(\mathcal{L}^{\otimes -1})_x$ are free $\mathcal{O}_{X,x}$-modules of rank 1. We conclude from Algebra, Nakayama's Lemma **??** that s_x is a basis for $\mathcal{L}_x$. Hence there exists a basis element $t_x \in (\mathcal{L}^{\otimes -1})_x$ such that $s_x \otimes t_x$ maps to 1. Choose an open neighbourhood U of x such that t_x comes from a section t of $\mathcal{L}^{\otimes -1}$ over U and such that $s \otimes t$ maps to $1 \in \mathcal{O}_X(U)$. Clearly, for every $x' \in U$ we see that s generates the module $\mathcal{L}_{x'}$. Hence $U \subset X_s$. This proves that X_s is open. Moreover, the section t constructed over U above is unique, and hence these glue to give the section s' of the lemma. ☐

Recall from Modules, Lemma **??** that given an invertible sheaf $\mathcal{L}$ on a locally ringed space X, and given a global section s of $\mathcal{L}$ the set $X_s = \{x \in X \mid s \notin \mathfrak{m}_x\mathcal{L}_x\}$ is open. A general remark is that $X_s \cap X_{s'} = X_{ss'}$, where ss' denote the section $s \otimes s' \in \Gamma(X, \mathcal{L} \otimes \mathcal{L}')$.

Definition 10.2. Let X be a scheme. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module. We say $\mathcal{L}$ is *ample* if

(1) X is quasi-compact, and
(2) for every $x \in X$ there exists an $n \geq 1$ and $s \in \Gamma(X, \mathcal{L}^{\otimes n})$ such that $x \in X_s$ and X_s is affine.

Exercise 10.3. Let L be an ample bundle on a K3 surface M. Prove that $\mathcal{L}^{\otimes 2}$ is globally generated (that is, for each $x \in M$ there exsits a section $h \in H^0(L^{\otimes 2})$ which does not vanish in x).

Proof. This just asks that the n in Definition 10.2 is 2 for all $x \in X$. Because, again, $x \in X_s$ means that $s(x) \neq 0$ because if it was, then we could somehow write s as a product of a vanishing function on $\mathfrak{m}_x$ and a local frame of $\Gamma(X, \mathcal{L})$. But I guess for the exercise do this: a line bundle is *ample* if there is n such that the canonical embedding (cf Lemma 10.5) is an embedding, i.e. that $\mathcal{L}^{\otimes n}$ is *very ample.* (Interestingly, the notion very ampleness is defined in morphisms.tex.) ☐

Now we pass to the part where ampleness gives you an **open immersion** to some projective space. Because, it's only very ampleness that gives an embedding, right? (Actually I think here in stacks project there are no embeddings but closed immersions.)

Definition 10.4. From modules.tex. Let $(X, \mathcal{O}_X)$ be a ringed space. Given an invertible sheaf $\mathcal{L}$ on X we define the *associated graded ring* to be

$$\Gamma_*(X, \mathcal{L}) = \bigoplus\nolimits_{n \geq 0} \Gamma(X, \mathcal{L}^{\otimes n})$$

Given a sheaf of $\mathcal{O}_X$-modules $\mathcal{F}$ we set

$$\Gamma_*(X, \mathcal{L}, \mathcal{F}) = \bigoplus\nolimits_{n \in \mathbf{Z}} \Gamma(X, \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes n})$$

which we think of as a graded $\Gamma_*(X, \mathcal{L})$-module.

Lemma 10.5. *Let X be a scheme. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module. Set $S = \Gamma_*(X, \mathcal{L})$ as a graded ring. If every point of X is contained in one of the open subschemes X_s, for some $s \in S_+$ homogeneous, then there is a canonical morphism of schemes*

$$f : X \longrightarrow Y = \mathit{Proj}(S),$$

to the homogeneous spectrum of S (see Constructions, Section **??***). This morphism has the following properties*

1. *$f^{-1}(D_+(s)) = X_s$ for any $s \in S_+$ homogeneous,*
2. *there are $\mathcal{O}_X$-module maps $f^*\mathcal{O}_Y(n) \to \mathcal{L}^{\otimes n}$ compatible with multiplication maps, see Constructions, Equation (***??***),*
3. *the composition $S_n \to \Gamma(Y, \mathcal{O}_Y(n)) \to \Gamma(X, \mathcal{L}^{\otimes n})$ is the identity map, and*
4. *for every $x \in X$ there is an integer $d \geq 1$ and an open neighbourhood $U \subset X$ of x such that $f^*\mathcal{O}_Y(dn)|_U \to \mathcal{L}^{\otimes dn}|_U$ is an isomorphism for all $n \in \mathbf{Z}$.*

11. Riemann-Roch theorem for curves

Let $D = \sum n_p p$ be a divisor on a curve. How many rational functions $z \in K^*$ satisfy $\mathrm{div}(z) \geq -D$? That is, rational functions which can only have poles of order $\leq n_p$ at the points p with $n_p > 0$, and can only have zeros of order $\geq |n_p|$ at the points p with $n_p < 0$.

For example, fix a point p in the curve. Then $\mathcal{O}_C(p)$ is the space of rational functions with a simple pole at p. $\mathcal{O}_C(2p)$ is the space of rational functions with a pole of order at most 2 at p, and so on. Note that $\mathcal{O}_C(p) \subset \mathcal{O}_X(2p) \subset \mathcal{O}(3p) \ldots$. That is, the more poles we allow, this space of functions will be larger.

In classical (Riemann's) notation, this space was denoted $L(D)$, and its dimension $\ell(D)$. Modern notation reads $H^0(\mathcal{O}_C(D))$ instead of $L(D)$ and $h^0(\mathcal{O}_C(D))$ instead of $\ell(D)$.

Riemann-Roch formula for curves is

$$h^0(D) - h^0(K - D) = \deg(D) - g + 1, \tag{11.0.1}$$

where K is a divisor associated to the curve canonically. This is done by choosing a meromorphic section s of the bundle Ω^1_C and taking $K := \mathrm{div}(s)$. Then one shows that this is independent of the choice of s.

Note that by Serre duality $h^1(D) = h^0(K - D)$.

12. Schemes

For a definition of presheaf, see Categories, Definition **??**.

Definition 12.1. Let X be a topological space.

(1) A *sheaf* $\mathcal{F}$ *of sets on* X is a presheaf of sets which satisfies the following additional property: Given any open covering $U = \bigcup_{i\in I} U_i$ and any collection of sections $s_i \in \mathcal{F}(U_i)$, $i \in I$ such that $\forall i, j \in I$
$$s_i|_{U_i\cap U_j} = s_j|_{U_i\cap U_j}$$
there exists a unique section $s \in \mathcal{F}(U)$ such that $s_i = s|_{U_i}$ for all $i \in I$.

(2) A *morphism of sheaves of sets* is simply a morphism of presheaves of sets.

(3) The category of sheaves of sets on X is denoted $Sh(X)$.

Let X be a topological space. Let $x \in X$ be a point. Let $\mathcal{F}$ be a presheaf of sets on X. The *stalk of* $\mathcal{F}$ *at* x is the set
$$\mathcal{F}_x = \operatorname{colim}_{x\in U} \mathcal{F}(U)$$
where the colimit is over the set of open neighbourhoods U of x in X. The set of open neighbourhoods is partially ordered by (reverse) inclusion: We say $U \geq U' \Leftrightarrow U \subset U'$. The transition maps in the system are given by the restriction maps of $\mathcal{F}$. See Categories, Section **??** for notation and terminology regarding (co)limits over systems. Note that the colimit is a directed colimit. Thus it is easy to describe $\mathcal{F}_x$. Namely,
$$\mathcal{F}_x = \{(U, s) \mid x \in U, s \in \mathcal{F}(U)\}/\sim$$
with equivalence relation given by $(U, s) \sim (U', s')$ if and only if there exists an open $U'' \subset U \cap U'$ with $x \in U''$ and $s|_{U''} = s'|_{U''}$. Given a pair (U, s) we sometimes denote s_x the element of $\mathcal{F}_x$ corresponding to the equivalence class of (U, x). We sometimes use the phrase "image of s in $\mathcal{F}_x$" to denote s_x. For example, given two pairs (U, s) and (U', s') we sometimes say "s is equal to s' in $\mathcal{F}_x$" to indicate that $s_x = s'_x$. Other authors use the terminology "germ of s at x".

Now let's discuss closed immersions.
Closed immersions of ringes spaces are in 01C1.

Definition 12.2. A map of ringed spaces $i : (Z, \mathcal{O}_Z) \to (X, \mathcal{O}_X)$ is a *closed immersion* if i is injective, its image is closed (I think these two are "closed immersion of topological spaces") and the induced map $\mathcal{O}_X \to i_*\mathcal{O}_Z$ is surjective; denote its kernel by $\mathcal{I}$. To make sure that if $(X, \mathcal{O}_X)$ is a scheme then so is $(Z, \mathcal{O}_Z)$ with add the extra condition that the $\mathcal{O}_X$-module $\mathcal{I}$ is locally generated by sections.

Now let's discuss restricting sheaves to subschemes.

Upshot about restricting sheaves to subschemes: the sheaves on Z correspond essentially (in the categorical sense) to sheaves on X with support on Z.

More explicitly, from 01QY: the pushforward of the inclusion (a closed immersion of schemes) $i : Z \to X$, with kernel on induced map $\mathcal{O}_Z \to \mathcal{O}_X$ denoted by I, is an exact, fully faithful functor
$$i_* : QCoh(\mathcal{O}_Z) \to QCoh(\mathcal{O}_X)$$
with essential image the quasi-coherent $\mathcal{O}_X$ modules $\mathcal{G}$ such that $\mathcal{I}\mathcal{G} = 0$.

It makes sense to think of this operation as $\mathcal{F}|_Z = i^*\mathcal{F} = \mathcal{F} \otimes \mathcal{O}_Z$ for $\mathcal{F} \in QCoh(X)$. Indeed: tensoring by the ideal sheaf would kill the information on Z, and tensoring with the structure sheaf does the opposite — it's the sheaves on X that are no fun outside Z, so basically sheaves on Z.

Here's a more detailed discussion:

In understanding the cohomology of sheaves when we restrict them to submanifolds (see Complex Geometry Exercise 22.3, I found Stacks Project 01AW section on closed immersions and abelian sheaves. First we have a few propositions stating that sheaves on the subscheme $Z \subset X$ can be pushed to sheaves on X.

And then there's the converse: what I would like to call the restriction functor $\mathcal{H}$ but rather is called the *sub-module of sections with support in* Z. In Remark 01AY (and then in Remark 0G6N for ringed spaces; these seem to be analogue constructions...) we see that for an abelian sheaf $\mathcal{F}$ on X we can define an abelian sheaf $\mathcal{H}_Z(\mathcal{F})$ which although is a sheaf on X we can just think it's a sheaf on Z - that's beacause

$$\mathcal{H}_Z(\mathcal{F})(U) = \{s \in \mathcal{F}(U))|\text{the support of } s \text{ is contained in } Z \cap U\}.$$

We can just think that it's the functions that vanish outside the subvariety. Which doesn't really make sense: a function vanishing outside the submanifold will most likely vanish also in the submanifold! Just by continuity, the submanifold is in the closure of the open set! I guess what I mean to say is that I don't mind if the functions do not vanish outside Z... I just want to consider them as functions on Z...

Here's another attempt in understanding how to restrict sheaves to submanifolds. First and foremost, lacking a reference this, I understand the symbol $\mathcal{F}|_Z$ for a sheaf on X and a subscheme $Z \subset X$ to mean the pullback of $\mathcal{F}$ by the inclusion. Then look at 01QY (more basically, as above, 01AW... in fact there appear to be several version of this statement) to recall that sheaves of the subscheme Z should correspond to sheaves on X with support on Z. **So essentially $\mathcal{F}|_Z$ should be a version of $\mathcal{F}$ but with support on Z.** The key point here is that this is non other than $\mathcal{F} \otimes \mathcal{O}_Z$. Why? Because when multiply with the ideal sheaf of Z we obtain $(\mathcal{F} \otimes \mathcal{O}_Z)\mathcal{I} = 0$ since sections of $\mathcal{I}$ vanish in $\mathcal{O}_Z = \mathcal{O}_X/\mathcal{I}$. That is, the support of $\mathcal{F} \otimes \mathcal{O}_Z$ indeed contained in Z: if the sections become zero when multiplied by sections of $\mathcal{I}$ (sections of $\mathcal{I}$ are only nonzero outside Z) it means that all the action was inside Z. that outside Z

I think the upshot here is that structure sheaves act as some sort of "contour" for the geometric object: they vanish *outside* the geometric object, they tell us what is the shape of the object by describing its surroundings, rather than its interior. This is exactly how algebraic/analytic sets are defined, and this why, when we want to restrict a sheaf to a subscheme, we tensor with the structure sheaf: tensor by the ideal sheaf would kill the information on Z, and tensoring with the structure sheaf does the opposite.

Now let's dicuss Proj of a graded ring.
(Stacks Project 01M3, Proj of a graded ring.) This section may be taken as the preamble to the question: what is projective space anyway? Yes, projective space is Proj of a graded ring S. And yes, projective space is a scheme. So in particular it's locally affine.

(Stacks Project 01MX, Functoriality of Proj.) This is reminiscent of the olden days. The point is that Proj is covariant, i.e. for rings $S' \subset S$ we have Proj$S \subset$

Proj$S$ (under the right conditions, which is the point of this section), meaning that, indeed, **Projs are projective schemes**.

Noe let's discuss invertible sheaves on Proj (twists!).
As for the construction of the twisted sheaves: for a projective space Proj$S$, we just compute the twisted S module $S(m)$ by the formula $S(m)_d = S_{m+d}$ and turn that into a sheaf $\widetilde{S(m)}$.

Here's the construction of the twisted sheaves $\mathcal{O}_X(n)$. The point is that there is a good way to pass from an S-module M to an $\mathcal{O}_X$-module, called $\widetilde{M}$. So basically you just define the twists by $M(d)_n = M_{n+d}$ and apply that construction.

Pick an element of degree d. Then think of the module generated by this element. The least degree piece of such a module is the degree d piece of the original module, that is we have $M(-d)$.

Lost note on logarithmic derivations:
Just to record a notion of logarithmic derivation from Vladimiro Benedetti, Danielle Faenzi and Simone.

Let F be a homogneous polynomial of degree d. The *logarithmic derivations* of F are

$$\mathrm{Der}_U(-\log F) = \{\partial \in \mathrm{Der}_U : \partial F \in \langle F \rangle_U\}$$

where $U = \mathbb{C}[x_1, \dots, x_e]$.

Lost notes on gonality.
From a talk by Juliana Coelho at Bandoleiros 2025:

A curve C is *k-gonal* if there is a map $C \to \mathbb{P}^1$ of degree k.

A curve is *stable* if it is nodal and $\mathrm{Aut}(C)$ is finite. Equivalently, the rational components of C meet the rest of the curve in at least three points.

We may study the moduli of k-stable curves, or the space of k-gonal structures.

13. Reduced schemes

So far a reduced scheme, for me, is just a scheme where there are no "repetitions": (x^2) is not reduced because it gives the same information as (x). Algebraically this means that the coordinate ring is reduced, which is defined by asking that there are no nilpotent elements, i.e. elements such that some power vanishes. (There's no definition of reduced ring in Stacks Project since it's taken as part of the basic algebraic knowledge).

The definition is on stalks but the first lemma shows it's equivalent to define it on any open set. This is in virtue of some "injectivity" lemma from the ring at U to the product of all stalks parametrized in U; essentially that if a section vanishes when projected to all the stalks then it's zero.

Definition 13.1. Let X be a scheme. We say X is *reduced* if every local ring $\mathcal{O}_{X,x}$ is reduced.

Proof. Assume that X is reduced. Let $f \in \mathcal{O}_X(U)$ be a section such that $f^n = 0$. Then the image of f in $\mathcal{O}_{U,u}$ is zero for all $u \in U$. Hence f is zero, see Sheaves, Lemma **??**. Conversely, assume that $\mathcal{O}_X(U)$ is reduced for all opens U. Pick any nonzero element $f \in \mathcal{O}_{X,x}$. Any representative $(U, f \in \mathcal{O}(U))$ of f is nonzero and hence not nilpotent. Hence f is not nilpotent in $\mathcal{O}_{X,x}$. $\square$

14. Flat morphisms

The essential technical property for for defining flatness is the preservation of exact sequences. Right-exactness is true in general; it follows from currying in category of commutative rings. But the functor $-\otimes_R N$ does not preserve injectivity of maps— that's the point of flatness.

Lemma 14.1 (Internal Hom for R-modules)**.** *For any three R-modules M, N, P,*

$$\mathrm{Hom}_R(M \otimes_R N, P) \cong \mathrm{Hom}_R(M, \mathrm{Hom}_R(N, P))$$

Proof. An R-linear map $\hat{f} \in \mathrm{Hom}_R(M \otimes_R N, P)$ corresponds to an R-bilinear map $f : M \times N \to P$. For each $x \in M$ the mapping $y \mapsto f(x, y)$ is R-linear by the universal property. Thus f corresponds to a map $\phi_f : M \to \mathrm{Hom}_R(N, P)$. This map is R-linear since

$$\phi_f(ax + y)(z) = f(ax + y, z) = af(x, z) + f(y, z) = (a\phi_f(x) + \phi_f(y))(z),$$

for all $a \in R$, $x \in M$, $y \in M$ and $z \in N$. Conversely, any $f \in \mathrm{Hom}_R(M, \mathrm{Hom}_R(N, P))$ defines an R-bilinear map $M \times N \to P$, namely $(x, y) \mapsto f(x)(y)$. So this is a natural one-to-one correspondence between the two modules $\mathrm{Hom}_R(M \otimes_R N, P)$ and $\mathrm{Hom}_R(M, \mathrm{Hom}_R(N, P))$. □

Lemma 14.2 (Tensor product is right exact)**.** *Let*

$$M_1 \xrightarrow{f} M_2 \xrightarrow{g} M_3 \to 0$$

be an exact sequence of R-modules and homomorphisms, and let N be any R-module. Then the sequence

$$M_1 \otimes N \xrightarrow{f\otimes 1} M_2 \otimes N \xrightarrow{g\otimes 1} M_3 \otimes N \to 0 \tag{14.2.1}$$

is exact. In other words, the functor $-\otimes_R N$ is right exact, in the sense that tensoring each term in the original right exact sequence preserves the exactness.

Proof. For every R-module P we apply the functor $\mathrm{Hom}(-, \mathrm{Hom}(N, P))$ to the first exact sequence. We obtain

$$0 \to \mathrm{Hom}(M_3, \mathrm{Hom}(N, P)) \to \mathrm{Hom}(M_2, \mathrm{Hom}(N, P)) \to \mathrm{Hom}(M_1, \mathrm{Hom}(N, P))$$

which is exact by Lemma **??** (1). By Lemma 14.1 this becomes the sequence

$$0 \to \mathrm{Hom}(M_3 \otimes N, P) \to \mathrm{Hom}(M_2 \otimes N, P) \to \mathrm{Hom}(M_1 \otimes N, P)$$

which is therefore also exact. Then using Lemma **??** (1) again, we arrive at the desired exact sequence. □

Remark 14.3. However, tensor product does NOT preserve exact sequences in general. In other words, if $M_1 \to M_2 \to M_3$ is exact, then it is not necessarily true that $M_1 \otimes N \to M_2 \otimes N \to M_3 \otimes N$ is exact for arbitrary R-module N.

Example 14.4. Consider the injective map $2 : \mathbf{Z} \to \mathbf{Z}$ viewed as a map of $\mathbf{Z}$-modules. Let $N = \mathbf{Z}/2$. Then the induced map $\mathbf{Z} \otimes \mathbf{Z}/2 \to \mathbf{Z} \otimes \mathbf{Z}/2$ is NOT injective. This is because for $x \otimes y \in \mathbf{Z} \otimes \mathbf{Z}/2$,

$$(2 \otimes 1)(x \otimes y) = 2x \otimes y = x \otimes 2y = x \otimes 0 = 0$$

Therefore the induced map is the zero map while $\mathbf{Z} \otimes N \neq 0$.

Definition 14.5. For R-modules N, if the functor $-\otimes_R N$ is exact, i.e. tensoring with N preserves all exact sequences, then N is said to be *flat* R-module. We will discuss this later in Section **??**.

Epiphany: in the category of commutative rings pushout is tensor product. So think of Spec as a functor from $CRing^{\mathrm{op}}$ to Sch, then pushout goes to pullback, and what's an example of a pullback? Fibre! So, the coordinate ring of a fiber is essentially given by the residue field at the point that parametrizes it! (tensored with the coordinate ring of the deformation space).

There is a lot of information on Stacks Project about flatness. It looks like the heart of the concept is captured in the commutative-algebraic notion of preserving exact sequences:

Definition 14.6. Let R be a ring.

(1) An R-module M is called *flat* if whenever $N_1 \to N_2 \to N_3$ is an exact sequence of R-modules the sequence $M \otimes_R N_1 \to M \otimes_R N_2 \to M \otimes_R N_3$ is exact as well.
(2) An R-module M is called *faithfully flat* if the complex of R-modules $N_1 \to N_2 \to N_3$ is exact if and only if the sequence $M \otimes_R N_1 \to M \otimes_R N_2 \to M \otimes_R N_3$ is exact.
(3) A ring map $R \to S$ is called *flat* if S is flat as an R-module.
(4) A ring map $R \to S$ is called *faithfully flat* if S is faithfully flat as an R-module.

Recall that a module M over a ring R is *flat* if the functor $-\otimes_R M : \mathrm{Mod}_R \to \mathrm{Mod}_R$ is exact. A ring map $R \to A$ is said to be *flat* if A is flat as an R-module.

15. Hilbert polynomial

The following lemma will be made obsolete by the more general Lemma **??**.

Lemma 15.1. *Let k be a field. Let $n \geq 0$. Let $\mathcal{F}$ be a coherent sheaf on $\mathbf{P}^n_k$. The function*

$$d \longmapsto \chi(\mathbf{P}^n_k, \mathcal{F}(d))$$

is a polynomial.

Proof. We prove this by induction on n. If $n = 0$, then $\mathbf{P}^n_k = \mathrm{Spec}(k)$ and $\mathcal{F}(d) = \mathcal{F}$. Hence in this case the function is constant, i.e., a polynomial of degree 0. Assume $n > 0$. By Lemma **??** we may assume k is infinite. Apply Lemma **??**. Applying Lemma **??** to the twisted sequences $0 \to \mathcal{F}(d-1) \to \mathcal{F}(d) \to i_*\mathcal{G}(d) \to 0$ we obtain

$$\chi(\mathbf{P}^n_k, \mathcal{F}(d)) - \chi(\mathbf{P}^n_k, \mathcal{F}(d-1)) = \chi(H, \mathcal{G}(d))$$

See Remark **??**. Since $H \cong \mathbf{P}^{n-1}_k$ by induction the right hand side is a polynomial. The lemma is finished by noting that any function $f : \mathbf{Z} \to \mathbf{Z}$ with the property that the map $d \mapsto f(d) - f(d-1)$ is a polynomial, is itself a polynomial. We omit the proof of this fact (hint: compare with Algebra, Lemma **??**). $\square$

Definition 15.2. Let k be a field. Let $n \geq 0$. Let $\mathcal{F}$ be a coherent sheaf on $\mathbf{P}^n_k$. The function $d \mapsto \chi(\mathbf{P}^n_k, \mathcal{F}(d))$ is called the *Hilbert polynomial* of $\mathcal{F}$.

The Hilbert polynomial has coefficients in $\mathbf{Q}$ and not in general in $\mathbf{Z}$. For example the Hilbert polynomial of $\mathcal{O}_{\mathbf{P}^n_k}$ is

$$d \longmapsto \binom{d+n}{n} = \frac{d^n}{n!} + \ldots$$

This follows from the following lemma and the fact that

$$H^0(\mathbf{P}^n_k, \mathcal{O}_{\mathbf{P}^n_k}(d)) = k[T_0, \ldots, T_n]_d$$

(degree d part) whose dimension over k is $\binom{d+n}{n}$.

Lemma 15.3. *Let k be a field. Let $n \geq 0$. Let $\mathcal{F}$ be a coherent sheaf on $\mathbf{P}^n_k$ with Hilbert polynomial $P \in \mathbf{Q}[t]$. Then*

$$P(d) = \dim_k H^0(\mathbf{P}^n_k, \mathcal{F}(d))$$

for all $d \gg 0$.

Proof. This follows from the vanishing of cohomology of high enough twists of $\mathcal{F}$. See Cohomology of Schemes, Lemma **??**. □

For completeness I include earlier notes from [Har77] on the matter.

The fact that M is finitely generated is what makes the following two definitions make sense.

Definition 15.4. The *Hilbert function* of a finitely generated graded $S = k[x_0, \ldots, x_r]$ -module M is

$$H_M(d) = \dim_k M_d$$

Definition 15.5. Define F_0 to be the free S-module on the generators of M. Elements in the kernel M_1 of the inclusion are called *sysygies.* By Hilbert's basis theorem, M_1 is also finitely generated, so we may choose a set of generators and repeat this process.

Theorem 15.6 (Hilbert Syzygy Theorem)**.** *Any finitely generated S-module M has a finite graded free resolution*

$$0 \longrightarrow F_m \xrightarrow{\varphi_m} \quad \longrightarrow F_{m-q} \longrightarrow \cdots \longrightarrow F_1 \xrightarrow{\varphi_1} F_0$$

Moreover, we may take $m \leq r+1$, the number of variables in S.

Lemma 15.7. *Suppose that $S = k[x_0, \ldots, x_r]$ is a polynomial ring. If the graded S-module M has finite free resolution*

$$0 \longrightarrow F_m \xrightarrow{\varphi_m} F_{m-1} \longrightarrow \cdots [r] \qquad F_1 \xrightarrow{\varphi_1} \quad \longrightarrow F_0$$

with each F_i a finitely generated free module, $F_i = \bigoplus_j S(-a_{i,j})$, then

$$H_M(d) = \sum_{i=0} (-1)^i \sum_j \binom{r+d-a_{i,j}}{r} \tag{15.7.1}$$

Lemma 15.8. *There is a polynomial $P_M(d)$ called the Hilbert polynomial such that, if M has free resolution as above, then $P_M(d) = H_M(d)$ for $d \geq max_{i,j}\{a_{i,j} - r\}$.*

Proof. When d satisfies this bound then the binomial coefficients in Eq. 15.7.1 are polynomials of degree r in d. □

Theorem 15.9 (Hilbert-Serre)**.** *Let M be a finitely generated graded $S = k[x_0, \dots, x_n]$. Then there exists a unique polynomial p_M such that $p_M(\ell) = \dim S_\ell$ for large enough ℓ.*

Definition 15.10. The polynomial P_M of Hilbert-Serre Theorem [**?**] is the *Hilbert polynomial* of the finitely generated $k[x_0, \dots, x_n]$-module M.

Definition 15.11. If $Y \subset \mathbb{P}^n$ is an algebraic set of dimension r, we define the *degree of* Y to be $r!$ times the leading coefficient of the Hilbert polynomial of the homogeneous coordinate ring $S(Y)$.

Exercise 15.12. Let H be a very ample divisor on the surface X, corresponding to a projective embedding $X \subseteq \mathbb{P}^N$. If we write the Hilbert polynomial of X as $P(z) = \frac{1}{2}az^2 + bz + c$, show that $a = H^2$, $b = \frac{1}{2}H^2 + 1 - \pi$, where π is the genus of a nonsingular curve representing H, and $c = 1 + p_a$.

16. Nakai-Moishezon Criterion

Theorem 16.1 (Nakai-Moishezon Criterion)**.** *A divisor D on the surface X is ample if and only if $D^2 > 0$ and $D.C > 0$ for all irreducible curves C in X.*

Proof. The direct implication is easy: since D is ample, mD is very ample for some m, so that m^2D^2 is the self-intersection number of mD. By exercise 15.12, D^2 is the leading coefficient of the Hilbert polynomial of X as a subscheme of $\mathbb{P}^n$. This means that D^2 is twice the leading coefficient of the Hilbert polynomial of a projective variety for large enough m, so that it must be a positive number (it's the dimension of one of the graded components of the coordinate ring of the surface). $\square$

17. Hilbert scheme

Upshot [**?**, p. 6]**.** We wish to parametrize subschemes of a projective space (or perhaps a more general scheme?). Since there are too many such subschemes we restrict ourselves to schemes with a given Hilbert polynomial, since the latter "encodes the most important numerical invariants of schemes". The Hilbert scheme is introduced via a theorem by Grothendieck as the object that represents the functor $\mathbf{Hilb}_{P,r}$ that maps a reduced scheme B to the set of proper flat families

$$\begin{tikzcd} \mathcal{X} \arrow[r, hook, "i"] \arrow[dr] & \mathbb{P}^r \times B \arrow[d, "\pi_B"] \\ & B \end{tikzcd}$$

with $\mathcal{X}$ having Hilbert polynomial P.

Theorem 17.1 (Grothendieck, '66)**.** *The functor $\mathbf{Hilb}_{P,r}$ is representable by a projective scheme $\mathcal{H}_{P,r}$.*

SEE Hilbert schemes of subschemes.

18. Deformation theory

[**?**] has a great way of motivating deformation theory via the moduli problem. Here let's just put the following important meaning for "deformation theory" (as opposed to studying flat families!):

> "*Deformation theory* is the study of infinitesimal deformaitons as a tool to understand the local structure of the moduli space. The goal is to be able to describe the restriction of the universal family to a small neighborhood of $m \in \mathcal{M}$, or, more precisely, its restriction to the germ of M at m.
>
> What is interesting here is that we can study order and infinitesimal deformations even though the functor F is not representable or simply we don't yet know it is. This is the most frequent case. Such an investigation will reveal the infinitesimal properties at $[X]$ of a yet unknown global structure on M which will be hopefully understood at a subsequent stage of the investigation. In order words it turns out to be possible and convenient to separate the *global moduli problem* from the *local moduli problem*, and deformation theory studies the latter, with the purpose of constructing a family of deformations of a given object parametrized by the spectrum of a local ring, and having properties as close as possible to a universal property."

I start by reading Stacks Project.

The first notion is *thickening of ringed spaces*, which I ultimately think of as a closed subscheme $(X, \mathcal{O}_X) \to (X', \mathcal{O}_{X'})$ with a nilpotent ideal sheaf, which in an imprecise way means that $\mathcal{I}_X^n = 0$ for some n, and in a precise way its given on sections.

The following definition is from [lucas-defos], which in turn comes from [Ser06]

Definition 18.1. Let X be an algebraic $\mathbb{C}$-scheme.

(1) A *deformation* of X is a Cartesian diagram ξ

$$\begin{tikzcd} X \arrow[r] \arrow[d] & \mathcal{X} \arrow[d, "\pi"] \\ \operatorname{Spec}\mathbb{C} \arrow[r, "s"] & S \end{tikzcd}$$

where π is a flat surjective morphism of algebraic $\mathbb{C}$-schemes and S is connected. (Recall that flatness accounts for "continuity".)

(2) A *local deformation* of X is a deformation ξ where $S = \operatorname{Spec} A$ for A a noetherian local $\mathbb{C}$ algebra with residue field $\mathbb{C}$.

(3) An *infinitesimal deformation of* X is a local deformation with A an artinian local $\mathbb{C}$-algebra with residue field $\mathbb{C}$. X is called *rigid* if all infinitesimal deformations are trivial.

(4) An *inifinitesimal deformation of order* n is an infinitesimal deformation when $S = \operatorname{Spec}(\mathbb{C}[t]/(t^{n+1})$.

Upshot. An interpretation of the so-called *dual numbers* $k[t]/(t^2)$ (see [Har09]) as the tangent space of something is thinking of Taylor polynomials: after quotienting by t^2 we loose the tails of the polynomials and are left with the first derivative information only - this justifies the use of the "first order deformations" term.

There is the following interpretation in [continued-fractions] p. 39: the space of first order deformation classes of X is $D(\mathbb{C}[t]/(t^2)$. This is said to ""represent" the tangent space $\mathbb{T}_X^1$ of the hypothetical deformation space of X". (I put double quotations because the word “represent” is quoted in the original text.) Further, if X is nonsingular and compact, then $\mathbb{T}^1_X = H^1(X, T_X)$.

Which basically I interpret as: the dimension of the deformation space of a smooth compact variety is $H^1(X, T_X)$.

After a few months of the previous text in this section I think I have understood the construction of Kodaira-Spencer map for embedded deformations, which is [Ser06, Proposition 3.2.1(ii)].

But I will actually briefly outline a particular case of this, which is easier: [Ser06, Examples 3.2.4(ii)]. This the case when $X \subset Y$ is a Cartier divisor. Then X is cut out locally by a set of functions f_i defined on an affine cover U_i of Y. Then we have a line bundle defined by the transition functions $f_{ij} = f_i/f_j$. This is the line bundle $\mathcal{O}_Y(X)$, which by the adjunction formula is also the normal bundle $N_{X/Y}$ of X in Y.

Now consider a first order deformation

$$\begin{tikzcd} X \arrow[r] \arrow[d] & \mathcal{X} \subset Y \times B_\varepsilon \arrow[d, "\pi"] \\ 0 \arrow[r] & B_\varepsilon \end{tikzcd}$$

The great thing about first order deformations is that the structure sheaf of $\mathcal{X}$ on an affine open U_i of Y looks like the tensor product $\mathcal{O}_{U_i} \otimes \mathbb{C}[\varepsilon]/(\varepsilon^2)$ (01I4). By definition of tensor product this just says that sections look like $f + \varepsilon g$ for $f, g \in \Gamma(\mathcal{O}_{U_i})$.

If $\mathcal{X}$ is a Cartier divisor on $Y \times B_\varepsilon$ (it is; but I'm missing a formal proof; the intuition is that an “infinitesimal thickening” $\mathcal{X}$ of X has the same underlying topological space but different scheme structure; 01JL 01I4), it is given as the vanishing locus of some local sections $f_i + \varepsilon g_i$. (Perhaps the f_i coincide with the distinguished section of X.) The line bundle associated to $\mathcal{X}$ is given by the cocycle

$$F_{ij} = f_{ij} + \varepsilon g_{ij}$$

for $g_{ij} = \frac{g_i - f_{ij} g_j}{f_j}$. Indeed, note that $\frac{1}{f_j + \varepsilon g_j} = \frac{1}{f_j} - \varepsilon \frac{g_j}{f_j^2}$ and compute $F_{ij} = (f_i + \varepsilon g_j)/(f_j + \varepsilon g_j)$.

Then

$$f_i + \varepsilon g_i = (f_{ij} + \varepsilon g_{ij})(f_j + \varepsilon g_j)$$

gives

$$g_i = f_{ij} g_j + g_{ij} f_j$$

and since f_j vanishes along X, we are left with $g_i = f_{ij} g_j$ which is how a section of the sheaf $N_{X/Y}$ is constructed by definition of sheaf.

Example 18.2. Fix $g \geq 2$. The dimension of the deformation space of a nonsingular projective curve X is $3g - 3$. This is “the dimension of the moduli space of curves of genus g”.

We can compute this number by Riemann-Roch formula on the bundle $-K_X$. Indeed, since X is a curve, $\Omega^1_X = K_X$ and thus $-K_X = T_X$. We get

$$\begin{aligned} h^0(-K) - h^0(K - (-K)) &= \deg(-K) - g + 1 \\ &= -2g + 2 - g + 1 \qquad \text{degree additive and } \deg K = 2g-2 \\ &= -3g + 3 \end{aligned}$$

Now by Serre duality $h^0(K-(-K)) = h^1(-K) = h^1(T_X)$, and it turns out that a Riemann surface of genus $g \geq 2$ has no holomorphic vector fields, so that $h^0(-K) = h^0(T_X) = 0$.

Exercise 18.3. Let C be a smooth genus g curve which can be embedded in a K3 surface M, and X the family of all deformations of C in M.

(1) Prove that $\dim X \leq g$.
(2) Let $\mathcal{X}_g$ be the space of all curves of genus g (smooth?) which can be possibly embedded to a K3 surface. Prove that each irreducible component Z of $\mathcal{X}_g$ satisfies $\dim_{\mathbb{C}} \leq g+19$. Deduce that there exists a compact complex curve which cannot be embedded in a K3 surface.

Exercise 18.4. Let C be a smooth curve embedded in a K3 surface X. Show that the dimension of the deformation space of C is $\leq g$.

Proof. The deformation space of a variety is the space of isomorphism classes of deformations as explained above. It turns out that there is a way to associate 1-cocyles of the tangent sheaf to deformations, so that in fact the deformation space Def_1 is isomorphic to $H^1(X, \mathcal{T}_X)$ for any variety X.

For our curve C we thus know that the dimension of the space of deformations (deformations not necessarily contained in X) is $h^1(\mathcal{T}_C)$. The family of deformations of C that are contained in X is the Hilbert scheme of curves with fixed Hilbert polynomial $P(t)$ after quotienting by $\mathbb{C}s$, where s is the section whose vanishing locus is C. This says that the number we are looking for is $h^0(X, \mathcal{O}(C)) - 1$.

Before showing that in fact $h^0(X, \mathcal{O}(C)) = 1+g$, let's explain why this Hilbert scheme construction actually works. Intuitively, any curve lying in the same component as C would be deformation-equivalent to C — at least we know that it's Hilbert polynomial is the same. But it's not clear what's the deformation involving these two (this question should be addressed!). Further, this doesn't explain Vladimiro's suggesion of "taking quotient by $\mathbb{C}s$". To me it looks like I would just compute the dimension of the tangent space of $\mathrm{Hilb}^P(M)$ at C.

Now I will show that $h^0(X, \mathcal{O}(C)) = 1+g$ (see [**?**, Lemma 1.2.1, Remark 1.2.2]).

Consider the ideal sheaf exact sequence twisted by $\mathcal{O}(C)$:

$$0 \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}(C) \longrightarrow \mathcal{O}_X(C) \otimes \mathcal{O}_C = \mathcal{O}(C)|_C \longrightarrow 0$$

By X being a K3 we know that $H^1(\mathcal{O}_X) = 0$, so that we have the short exact sequence in cohomology

$$0 \longrightarrow H^0(\mathcal{O}_X) \longrightarrow H^0(\mathcal{O}(C)) \longrightarrow H^0(\mathcal{O}(C)|_C) \longrightarrow 0$$

so that $h^0(\mathcal{O}(C)) = 1 + h^1(\mathcal{O}(C)|_C)$. A version of adjunction formula says $\omega_C \cong (K_X \otimes \mathcal{O}(C)|_C$, and using that $K_X = \mathcal{O}_X$ we obtain $h^0(\mathcal{O}(C)|_C) = h^0(\omega_C) = g$.

To

For the record I put other thoughts I went through in solving this exercise.

Recall from [**?**, p. 146] that the normal bundle $\mathcal{N}$ of a hypersurface of a smooth variety X satisfies $\mathcal{N}^\vee \cong \mathcal{O}_X(-C)|_C$. Taking duals we get that $\mathcal{N} \cong \mathcal{O}_X(C)|_C$.

By adjunction formula $2g-2 = (\mathcal{O}(C), \mathcal{O}(C))$. Applying Riemann-Roch to $\mathcal{O}(C)$ (which is by definition the dual of the ideal sheaf of C, which is a line bundle on X), we obtain that $\chi(\mathcal{O}(C)) = 2 + \frac{1}{2}(\mathcal{O}(C), \mathcal{O}(C))$. Then $\chi(\mathcal{O}(C)) = g+1$.

Now recall that $\chi(\mathcal{O}(C)) = h^0(\mathcal{O}(C) - h^1(\mathcal{O}(C)) + h^2(\mathcal{O}(C))$. By Serre duality and X being a K3 surface we see that $h^2(\mathcal{O}(C)) \cong h^0(\mathcal{O}(-C))$, which is the ideal sheaf of C. Any section of such a sheaf would vanish along C, and since X is compact we conclude there cannot be any such section.

Now we show that also $h^1(\mathcal{O}(C)) = 0$ to conclude that $h^0(\mathcal{O}(C)) = g + 1$.

We thus conclude that $h^0(\mathcal{O}(C))$

Since C is smooth we can use the normal exact sequence

$$0 \longrightarrow \mathcal{T}_C \longrightarrow \mathcal{T}_X|_C \longrightarrow \mathcal{N} \longrightarrow 0$$

Taking Euler characteristic we see that $\chi(\mathcal{T}_C) + \chi(\mathcal{N}) = \chi(\mathcal{T}_X|_C)$.

This means that we should be done once we compute $\chi(\mathcal{T}_X|_C)$. For this we can use Riemman-Roch formula for coherent sheaves on a curve, which tells us that

$$\deg(\mathcal{T}_X|_C) = \chi(\mathcal{T}_X|_C) - \mathrm{rk}(\mathcal{T}_X|_C) \cdot \chi(\mathcal{O}_C)$$

But of course we know that since C is a curve, by Serre duality we get that $\chi(\mathcal{O}_C) = 1 - g$. (Indeed: $h^1(\mathcal{O}_C) = h^0(\Omega^1_C) := p_a(C)$.) And now the question is what is the degree. Apparently this is just the first Chern class c_1 of the restricted tangent bundle. So what is it? And what is $h^0(\mathcal{T}_C)$ if the genus is 0 or 1, and that's it.

I know that $h^0(\mathcal{T}_X) = 0$ and $h^1(\mathcal{T}_X) = 20$ by X being a K3 surface, but I'm not sure what happens when we restrict to C.

If $g \geq 2$ we know that $h^0(\mathcal{T}_C) = 0$, so that $\boxed{\chi(\mathcal{T}_C) = h^1(\mathcal{T}_C)}$. To compute the latter Euler characteristic, which is given by definition by $\chi(\mathcal{T}_X|_C) = h^0(\mathcal{T}_X|_C) - h^1(\mathcal{T}_X|_C)$, we first note that $h^0(\mathcal{T}_X|_C) = 0$, because this is the dimension of the global holomorphic vector fields on X restricted to C, which is constant along C since X is smooth. And then this number is actually zero by Hodge numbers of a K3 surface.

and taking cohomology long exact sequence we obtain

$$\cdots \longrightarrow H^0(T_X) \longrightarrow H^0(\mathcal{N}) \longrightarrow H^1(T_C) \longrightarrow$$

$$H^1(T_X) \longrightarrow H^1(\mathcal{N}) \longrightarrow \cdots$$

Or is it $h^1(\mathcal{N})$? See [**?**]. If this was the case, then we can use adjunction formula as above to get that $2g - 2 = (\mathcal{N}, \mathcal{N}) = \deg_C \mathcal{N}$. Then we may find $h^0(\mathcal{N})$ via Riemann-Roch:

$$h^0(\mathcal{N}) - h^0(K_C - \mathcal{N}) = \deg \mathcal{N} - g + 1$$

Note that $h^0(N_C - \mathcal{N}) = h^1(-K_C + N + K_C) = h^1(\mathcal{N})$ via Serre duality, and by Riemann-Roch on a surface as above we see that

$$g + 1 = \chi(\mathcal{N}) = h^0(\mathcal{N}) - h^1(\mathcal{N}) \implies h^1(\mathcal{N}) = -g - 1 + h^0(\mathcal{N})$$

so that

$$h^0(\mathcal{N}) - (-g - 1 + h^0(\mathcal{N})) = \deg \mathcal{N} - g + 1$$

$$\implies \text{oops! I lost } h^0(\mathcal{N}) \text{ in this operation}\ldots$$

Maybe if I just use normal exact sequence and realise that

$$h^0(\mathcal{N}) = h^0(\mathcal{T}_X|_C) - h^0(\mathcal{T}_C)$$

I know that $h^0(\mathcal{T}_C) = 0$ for $g > 1$, so the question is how to compute the restricted holomorphic vector fields. $\square$

Look for the "Perfect obstruction theory". This is used by Bruzzo for deformations of (super)stacks, work with E. Paiva and —.

Until I have a better candidate of where to put this. . .

Definition 18.5. Let $A \to R$ be a ring homomorphism (i.e. R is an A-algebra). An *A-extension of R by I* is a short exact sequence

$$0 \longrightarrow I \longrightarrow E \xrightarrow{\varphi} R \longrightarrow 0.$$

where E is an R'-algebra and φ is a morphism of A-algebras whose kernel is I. We also require that $I^2 = (0)$; this condition implies that I has the sturcture of R-module.

Example 18.6.

$$0 \longrightarrow (\varepsilon) \longrightarrow A[\varepsilon] \longrightarrow A \longrightarrow 0$$

where $A[\varepsilon]$ is the algebra of dual numbers over A.

19. Cotangent modules

My aim in this section is to cover the minimal theory needed to apply results in [AC09] to my deformation problem. The main theorem I expect to use is the following:

Theorem 19.1. *If $\mathcal{K}$ is a 3-dimensional manifold then*

$$\dim T^1_{\mathbb{P}(K)} = 11d_3 + 5e_3 + 3e_4 + e_{\geq 5} + c_{\geq 6} + 5f_1^{(3)} + 2f_1^{(4)} + h^2(\mathcal{K}).$$

I need to do some interpretation first. From [AC08, p. 1]:

> "For each $\mathbb{C}$-algebra A, there is a cohomology theory providing modules T^i_A. However, only T^1_A and T^2_A are relevant for the deformation theory of $\operatorname{Spec} A$. The module T^1_A collects the infinitesimal deformations of A and T^2_A contains the obstructions for lifting these deformations to decent parameter spaces. Eventually, $\operatorname{Der}_{\mathbb{C}}(A, A)$ will be called T^0_A."

[AC08, p. 3]:

> "In general, for a finitely generated $\mathbb{C}$-algebra A, the modules T^i_A allow the following ad hoc definitions: Let $P = \mathbb{C}[x_0, \dots, x_n]$ mapping onto A so that $A \simeq P/I$ for an ideal I. The T^1_A is the cokernel of the natural map $\operatorname{Der}_{\mathbb{C}}(P, P) \to \operatorname{Hom}_P(I, A)$. Moreovr, [. . .] and we obtain T^2_A as the cokernel of the induced map $\operatorname{Hom}_P(P^m, A) \to \operatorname{Hom}_A(R/R_0, A)$."

Remark 19.2 (dani). Let's think more about that definition of T^1_A as the cokernel of the "natural map" $\operatorname{Der}_{\mathbb{C}}(P, P) \to \operatorname{Hom}_P(I, A)$.

Pick a derivation $\chi \in \operatorname{Der}_{\mathbb{C}}(P, P)$. We want a map $I \to A = P/I$. Pick an element $a \in I$ and map it to $\chi(a)$. And that's it. It might be zero or not. If the χ we picked maps any $a \in I$ to some element in I, then this χ is in the kernel of the natural map. So T^1_A is the image of the natural map modulo the kernel.

Unfortunately this doesn't say much about the following crucial fact: the T^i_A modules are $\mathbb{Z}^{n+1}$-graded. But here goes an explanation. We use the notation that e_p is a "face not in the complex", which of course corresponds to a generating monomial x^p of the SR ideal I. It has a degree, of course (given by the "support" of p). Now here's a rather sloppy step but: since the cotangent modules are built upon I, who is given by e_p, we just notice that the degree of e_p along with a (very natural, too) notion of degree among relations, induce a grading in all the cotangent modules T^i_A. Thus, the grading is morally just the degree of the monomials, which is in fact the support of p, a vector $\mathbf{c} \in \mathbb{Z}^{n+1}$.

Next in line is the decomposition $\mathbf{c} = \mathbf{a} - \mathbf{b}$ into positive minus negative part. But how exactly...?

20. Fano varieties

Definition 20.1. A *Fano variety* is a projective variety with $-K_X$ ample.

The idea of the Fano index is that the larger the index, the simpler is the variety. In a talk by Jõao, it is defined as the larger integer dividing $-K_X$ in $\operatorname{Pic}(X)$.

Definition 20.2.

$$r(X) := \min\{r : \frac{c_1(X)}{r} \in H^2(X, \mathbb{Z})\}$$

Exercise 20.3. By Kodaira vanishing theorem **??**, you can show that the cohomology $H^i(X, L)$ for a Fano variety X vanishes. You just have to put $L = \mathcal{O}(k)$ with $k \geq -r$, where r is the Fano index.

Exercise 20.4. Show that $\operatorname{Pic}(X) \cong H^2(X, \mathbb{Z})$ holds for Fano varieties.

Remark 20.5. If $H^3(X, \mathbb{Z}) = 0$ of a Fano 3-fold, then its derived category is generated by 4 elements.

Araujo-Druel em 2017 classify Fano foliations relating the Fano index (of the foliation) with its rank. João found conditions (integrability of the leaves) to describe del Pezzo foliations (a kind of Fano foliation) with Picard rank 1. Araujo (2019) classifies the leaves of a algebraically integrable del Pezzo foliation (with no restriction on the (log canonica) singularities!).

21. Quivers

Definition 21.1. A *quiver* is a set of vertices Q_0, a set of arrows Q_1 equipped with the maps of source s and target t that to each arrow they assign the point that is source or target of the arrow.

Definition 21.2. A *representation* of a quiver is a set of finite dimensional vector spaces equipped with maps between them realising a given quiver (incomplete...).

There is a notion of projective representation, which I missed to write. But it is analogous to the injective representation:

Definition 21.3. Given a quiver Q, the *injective representation* of Q_0 is given by, for $i \in Q_0$,

$$I(i)_j = \begin{cases} k & i = j \\ k^{d'} & j \neq i \end{cases}$$

where d' is the number of paths from j to i.

22. Stacks

My first definition of stack can be extracted from

Definition 22.1. A *superstack* is a stack over the étale site SSch of superschemes, i.e. it is a category fibered in groupoids over the category of superschemes, the latter equipped with the étale topology, satisfying the descent condition.

Here are some other definitions:

Definition 22.2. Let $\mathfrak{X}$ be a stack over $\mathrm{Sch}_{\text{ét}}$. An *algebraic space* is such that there exists morphism $\mathcal{U} \to \mathfrak{X}$ where $\mathcal{U}$ is a scheme, that is schematic, étale and injective (check this one).

$\mathfrak{X} \to y$ is *representable* if there exists a scheme $\mathcal{U}$ and a map $\mathcal{U} \to y$ such that the fibered product

$$\begin{tikzcd}
\mathcal{U} \times_y \mathfrak{X} \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \mathfrak{X} \arrow[d] \\
\mathcal{U} \arrow[r] & y
\end{tikzcd}$$

is an algebraic space.

Finally, a stack is *algebraic* (resp. *Deligne-Mumford*) is there exists a representable surjective morphism $\mathcal{U} \to \mathfrak{X}$ that is smooth (resp. étale).

A *stable map* over a projective variety X is an element of the first Chow group $\beta \in A_1$, where (C, g) is an algebraic curve and $f : C \to X$ with $[f(C)] = \beta$.

The curves that are points under this map (contractible) are **stable**.

23. Moduli

Pseudo-admissible covers may be the way Mumford orifinally developed stability notions.
This is a two-session minicourse by Ugo Bruzzo in LEGALzinho 2025.

Let X be a smooth complex projective variety, $X \overset{j}{\hookrightarrow} \mathbb{P}^n$. $\mathcal{O}_{\mathbb{P}^n}(1)$ is the class of hyperplanes of $\mathbb{P}^n$. $\mathcal{L} = j^*\mathcal{O}_{\mathbb{P}^n}(1)$. $c_1(\mathcal{L}) \in H^2(X, \mathbb{Z})$ is a *polarization*.

Let $\mathcal{E}$ be a coherent $\mathcal{O}_X$-module. For example, $\mathcal{E}$ may be a locally free $\mathcal{O}_X$-module. Recall that $H^k(X, \mathcal{E})$ are finite dimensional vector spaces and they vanish for $k > \dim X = n$. Recall that the *Euler characterstic* of $\mathcal{E}$ is given by $\chi(\mathcal{E}) = \sum_{k=0}^n (-1)^k \dim_{\mathbb{C}} H^k(X, \mathcal{E})$.

The *reduced Hilbert polynomial* of $\mathcal{E}$ is

$$p_{\mathcal{E}}(k) = \frac{1}{\mathrm{rk}\mathcal{E}} \chi(\mathcal{E} \otimes_{\mathcal{O}_X} \mathcal{L}^k).$$

Definition 23.1. Let $\mathcal{E}$ be a coherent $\mathcal{O}_X$-module with no torsion and positive rank. $\mathcal{E}$ is *stable* if for all $f \subset \mathcal{E}$ $(0\mathrm{rk} f < \mathrm{rk}\mathcal{E})$,

$$p_f(k) < p_{\mathcal{E}}(k) \qquad \text{for } k \gg 0$$

and *semi-stable* if we put $\leq$ instead of $<$.

Lemma 23.2. *If $\mathcal{E}$ is stable then $\mathcal{E}$ is irreducible $(\mathcal{E} \neq \mathcal{E}_1 \oplus \mathcal{E}_2)$ and $Aut(\mathcal{E}) \simeq \mathbb{C}^*$.*

The *functor of moduli* is

$$\begin{aligned}\mathcal{M} : \mathrm{Sch}_{\mathbb{C}} &\longrightarrow \mathsf{Set}\\ S &\longmapsto \left\{\begin{matrix}\text{stable family}\\ \text{of sheaves with}\\ \text{Hilbert polynomial} f\\ \text{parametrized by } S\end{matrix}\right\} \Big/ \sim .\end{aligned}$$

Riemann-Roch. Let C be a smooth, projective, connected curve of genus g. Let $\mathcal{E}$ be a coherent sheaf over $H^2(C,\mathbb{Z})$ with generator of $H^2(X,\mathbb{Z})$ $w = [pt] \in H^2(C,\mathbb{Z})$. $c_1(\mathcal{E}) = \deg\mathcal{E} \cdot w$.

Theorem 23.3.

$$\begin{aligned}\chi(\mathcal{E}) &= \int_C (1 + (1-g)w)(rk\mathcal{E} + deg\mathcal{E}w)\\ &= \int_C deg\mathcal{E} + w + (1-g)rk\mathcal{E} \cdot w\\ &= deg(\mathcal{E}) + 1 - grk\mathcal{E}.\end{aligned}$$

Example 23.4.

Theorem 23.5. *Se (r,d), $r > 0$ are coprime, the functor of moduli of stable bundles over C of rank r and degree d is representable, and the moduli space $\mathcal{M}_C(\mathcal{E},d)$ is smooth.*

In the non coprime case this does not necessarily hold.

Definition 23.6. $\mathcal{M}$ is a *grosseiro* moduli space if the funtor is not representable.

This means that although there is not a universal family, there is still a correspondence of the closed points of $\mathcal{M}$ and the equivalence classes of semistable sheaves of rank r and degree d.

Now let's fix $\dim X = 2$. Let $\mathcal{E}$ be a stable $\mathcal{O}_X$-module. Let $r = \mathrm{rk}\mathcal{E}$, and the *discriminant*

$$\Delta(\mathcal{E}) = c_2(\mathcal{E}) - \frac{\mathcal{E}-1}{2r} c_1(\mathcal{E})^2 \in H^4(X,\mathbb{Q}) \cong \mathbb{Q}.$$

Here enter $c_1(\mathcal{E}) \in H^2(X,\mathbb{Z})$ and $c_2(\mathcal{E}) \in H^4(X,\mathbb{Z})$.

Theorem 23.7 (Bogomolov)**.** $\Delta(\mathcal{E}) \geq 0$.

It could be interesting to study the proof of this theorem.

Miyaoka (1987). Consider a curve C and a bundle E. We can for any fibre, which is a vector space V, consider its projectivization $V \setminus \{0\}/\mathbb{C}^* = \mathbb{P}(V)$. Then we can bundlize this construction ad obtain $\mathbb{P}E \to C$. Consider $\mathcal{O}_{\mathbb{P}E}(1)$. We define

$$\lambda = rc_1(\mathcal{O}_{\mathbb{P}E}(1)) - \pi^* c_1(E) \in H^2(\mathbb{P}E,\mathbb{Z}).$$

Theorem 23.8. *E is semistable if λ is semicorrente efectiva.*

Definition 23.9. Consider again a projective smooth connected curve Γ and let $f : \Gamma \to \mathbb{P}E$. We say λ is *nef* if

$$\int_\Gamma f^*\lambda \geq 0.$$

The following theorem was proved by Nakayama (who is a different author than that of Nakayama's lemma) in 1999 but was unkown to Bruzo and Daniel HR.

Theorem 23.10 (Nakayama '99, Bruzzo-HR '04)**.** *The following are equivalent:*

(1) λ *is nef.*
(2) *For all* $f : \Gamma \to X$*,* f^*E *is semi-stable.*
(3) E *is semistable with respect to the polarization, and* $\Delta(E) = 0$*.*

The polarization says how the surface is embedded in projective space, it is $H = c_1(\mathcal{L})$. The stability notion is

$$\mu(E) = \frac{c_1(E) \cdot H^{n-1}}{\mathrm{rk}E}.$$

They have proved that for Higgs bundles $1 \iff 2$, $3 \implies 1, 2$. And also for $1, 2 \implies 3$ but only for $r = 2$ using the Higgs Grassmannian, but the problem is very complicated for general rank.

Second lecture. I think this might be a more general definition of the Hilbert polynomial:

$$\sum_{k=0}^{n} (-1)^k \dim H^k(X, \mathcal{E}) = \chi(\mathcal{E}) = \int_X \mathrm{ch}(\mathcal{E})\mathrm{td}X$$

And we also have:

$$\chi(\mathcal{E} \otimes \mathcal{L}^*) = \int_X (1 + (1-g)w)(r + kr \underbrace{c_1(\mathcal{L})}_{\mathrm{deg}\mathcal{L}\cdot w} + \mathrm{deg}\mathcal{E} \cdot w.$$

And then

$$p_{\mathcal{E}} = k\mathrm{deg}\mathcal{L} + 1 - g + \mu(\mathcal{E}).$$

Now we discuss Higgs bundles. Let X be ac onnected complex projective variety.

Definition 23.11. A *Higgs bundle* on X is a pair $\mathfrak{C} = (E, \phi)$ with

- E a vector bundle on X
- ϕ : a morphism (connection-like. . .)

For the following see Nitsuze, FGA explained. There following scheme is also representable (I think this is the "universal quotient")

We have also consider the Grassmanization of a vector bundle. We obtained a short exact sequence and tried to compute the tensor product with the holomorphic differentials. The idea was to pullback the grassman bundle given a map $f : Y \to X$. The result is not satisfactory, the pulled back thing may be strange. This is called the *Higgs Grassmannian*.

Here's some notion also studied by Campana, Demailly: numerically flat bnef bundle E. Then we have that

Lemma 23.12. *If* $c_1 = 0$ *then numerically flat bundle iff semistable.*

This is the beginning of a talk by Alan Muniz at Bandoleros 2025 titled "Rank 2-bundles on $\mathbb{P}^3$ with odd determinant". (Work in progress with A. Fontes.)

Problem: Classify rank-2 bundles on $\mathbb{P}^3$.

We consider bundles E with chern classes $c_1 = -1$, $c_2 = 2n$ and $c_3 = 0$. Let $\mathcal{B}(-1, 2n)$ be the moduli space of μ-stable rank 2-bundles.

- $\mathcal{B}(-1, 2)$ is irreducible of dimension 11 [Hartshorne-Sols '82, Manolache '81].
- $\mathcal{B}(-1, 4)$ has two components [Barnica-Manolache '85].

- $c_2 \geq 6$, Tikhomirov (2014) Fontes-Jardim (2023, 2024, 2025). $\mathcal{B}(-1,6)$ has ≥ 4 components.

Now let's discuss Serre correspondence.

How can we describe these spaces? Let E be a vector bundle of rank 2 on $\mathbb{P}^3$ and $\sigma \in H^0(E(r))$. Then $C = (\sigma_0) \subset \mathbb{P}^3$ is a curve * equipped with an "extension class" $\zeta \in H^0(\omega_C(4-2r-c_1))$. In fact, this correspondence is 1-1, and we call it Serre correspondence. We have that $\deg C = c_2(E(r))$ and $p_a(C) = 1 = \frac{c_2(E(r))(c_1(E(r))-4)}{2}$ To pass from the curve to the bundle and viceversa we use

$$0 \longrightarrow \mathcal{O}(-r) \longrightarrow E \longrightarrow \mathcal{I}_C(c_1+r) \longrightarrow 0$$

Upshot about Deligne-Mumford stacks: the sit below a stack via a map that is étale.

Now let's discuss framed moduli.
Framed instantons on the blow-up of $\mathbb{P}^3$ at a point.

Let X be a projective variety and fix a polarization H. Let $\mathcal{E}$ be a sheaf on X. A *framed pair* is a $(\mathcal{F},\alpha)$ where $\mathcal{F}$ is a coherent sheaf on X and α is a *framing datum*, i.e. a map $\alpha : \mathcal{F} \to \mathcal{E}$.

Set

$$\mathcal{E}(\alpha) = \begin{cases} 1 & \alpha \neq 0 \\ 0 & \text{otherwise.} \end{cases}$$

Define the Hilbert polynomial by

$$P(k) = \chi(\mathcal{F} \otimes \mathcal{O}(kH)).$$

For fied $\delta \in \mathbb{Q}[t]$ rational positive, define

$$P_{(\mathcal{F},\alpha)} = P_{\mathcal{F}}(k) - \varepsilon(\alpha)\delta.$$

We can also define induced framings from subsheaves $\mathcal{F}' \subset \mathcal{F}$, by asking that the inclusion commutes with the framing datum α. Then we put

$$\varepsilon(\alpha') = \begin{cases} 1 & \mathcal{F}' \subseteq Ker\alpha \\ 0 & \text{otherwise.} \end{cases}$$

We say that $\mathcal{F}$ is δ-*(semi)stable* if for any subpair $(\mathcal{F}',\alpha')$ with $\mathrm{rk}\mathcal{F}' = r' < r = \mathrm{rk}\mathcal{F}$ one has

$$\frac{1}{r'}P_{(\mathcal{F}',\alpha')}(k) \leq \frac{1}{r}P_{(\mathcal{F},\alpha)}(k).$$

If $\deg(\delta) \leq \dim X$, there are moduli functors $\mathcal{M}^{st}(X,\mathcal{E},\delta) \subseteq \mathcal{M}^{\mathrm{ss}}(X,\mathcal{E},\delta)$.

Remark 23.13. If $\deg\delta > \dim X$ then α is injective or zero, and $\mathcal{F}$ is semistable. This gives quot schemes.

Huybrechts-L. In $\dim X \leq 2$ there exists a quasi-projective fine moduli space for $\delta \in \mathbb{Q}[t]$ and $p \in \mathbb{Q}[k]$.

Then there is a notion of good framing by Bruzz-M, depending on D, an effective divisor on X. They proved that having good framing is enough to obtain a fine quasi-projective moduli space.

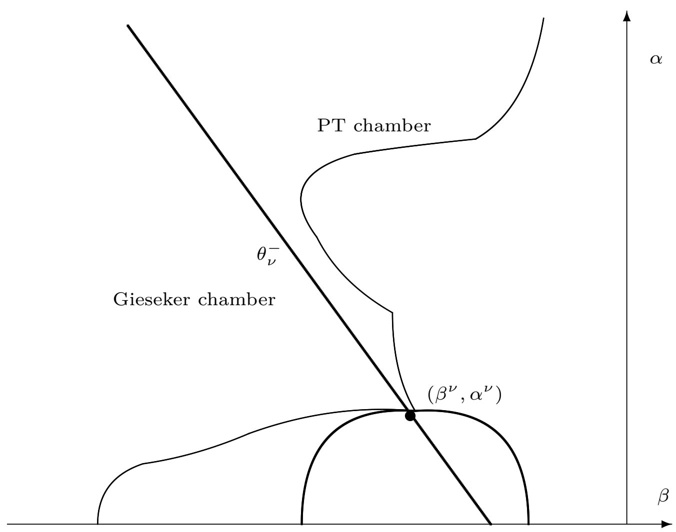
Definition 23.14. A coherent sheaf $\mathcal{E}$ on $X = \tilde{\mathbb{P}^3}$ is an *instanton* if it is $\mu_{\mathcal{L}}$-semistable, $c_1(\mathcal{E}) = 0$ and $H^0(\mathcal{E}) = 0 = H^0(Ec(-3,2))$, $H^1(\mathcal{E}(-2,1)) = 0 = H^2(\mathcal{E}(-2,1))$ and $H^3(\mathcal{E}(0,-1)) = H^2(\mathcal{E}(-1,1))$.

Some examples are locally free sheaves of rank 2 (via Harrshorne-Serre correspondence), torsion-free rank 2 constructed via elementary transformations.

General result: for rank 2 instantons on Fano threefolds that have a plane E, $\mathcal{E}|_E$ is μ-semistable if and only if $h^1(\mathcal{E}(-2,1) \otimes \mathcal{O}(-E)) = 0$.

24. Stability

Stability helps us select the building blocks of an abelian category, every object in the category can be reconstructed from those. Wall and chamber structure: inside a compact set $K \subset \mathrm{Stab}(\mathcal{T})$, only finitely many walls appear. If C is a connected component of the complement of the walls in K, then the stable objects are constant on C. Think of an ant walking in $\mathrm{Stab}(\mathcal{T})$ with a backpack of objects and a detector: when it crosses a wall and some object stops being stable, the red light fires. Stable objects change only when you cross a wall.



Let $\mathcal{A}$ be an abelian category with Grothendieck group $K_0(\mathcal{A})$.

Definition 24.1. An additive homomorphism $Z : K_0(\mathcal{A}) \to \mathbb{C}$ is a *stability function* if for all $0 \neq E \in \mathcal{A}$,

$$\mathrm{Im}Z(E) \geq 0 \qquad \text{and} \qquad \mathrm{Re}Z(E) < 0 \text{ if } \mathrm{Im}Z(E) = 0.$$

The *slope* of $E \in \mathcal{A}$ with respect to Z is

$$M(E) = \begin{cases} -\frac{\mathrm{Re}Z(E)}{\mathrm{Im}Z(E)} & \text{if } \mathrm{Im}Z(E) \neq 0 \\ \infty & \text{if } \mathrm{Im}Z(E) = 0. \end{cases}$$

A nonzero object $E \in \mathcal{A}$ is called *stable* if for all proper non trivial subobjects $F \subset E$, we have

$$M(F) < M(E)$$

and *semistable* if we put $\leq$ instead.

Definition 24.2. A pair $(\mathcal{A}, Z)$ is called a *stability condition* if

(1) Z is a stability function, and

(2) every $0 \neq E \in \mathcal{A}$ has a Harder-Narasimhan filtration.

Lemma 24.3. *Let $Z : K_0(\mathcal{A}) \to \mathbb{C}$ be a stability function. Assume that*

(1) *$\mathcal{A}$ is Noetherian, and*
(2) *the image of ImZ is discrete in $\mathbb{R}$.*

Then every $0 \neq E \in \mathcal{A}$ has a Harder-Narasimhan filtration with respect to Z.

Let $\mathcal{D}$ be a triangulated category. Fix a finite rank lattice Λ, a surjective homomorphism $v : K_0(\mathcal{D}) \to \Lambda$ and a norm $\| \cdot \|$ in Λ.

Definition 24.4. A *Bridgeland stability condition* $\mathcal{D}$ is a pair $(\mathcal{A}, Z)$ where $\mathcal{A}$ is the heart of a bounded t-structure on $\mathcal{D}$, such that

(1) $(\mathcal{A}, Z)$ is a stability condition,
(2) the following support property holds:

$$\inf \left\{ \frac{|Z(E)|}{\|v(E)\|} : 0 \neq E \in \mathcal{A}\text{semistable} \right\} > 0.$$

Example 24.5. Let X be a smooth projective curve and $\mathcal{D} = D^b(X)$. Take $\mathcal{A} = \mathcal{C}oh(X)$ as the heart of the standard t-structure. Let

$$Z(E) = -\deg(E) + i\,\mathrm{rk}(E).$$

Then σ-stability coincides with slope stability.

Let $Stab_\Lambda^{Br}(\mathcal{D})$ be the set of Bridgeland stability conditions and let $Stab_\Lambda(\mathcal{D})$ be the set of stability conditions on $\mathcal{D}$ with respect to (Λ, v).

Theorem 24.6 (Bridgeland)**.** *The natural map*

$$Stab_\Lambda(\mathcal{D}) \to \mathrm{Hom}(\Lambda, \mathbb{C})$$

is a local homeomorphism. $Stab_\Lambda(\mathcal{D})$ is a complex manifold and its dimension is the rank of Λ. In particular,

$$Stab_\Lambda^{Br}(\mathcal{D}) \subset Stab(\mathcal{D}).$$

Let $(\mathcal{A}, Z)$ be a stability condition and M be the slope with respect to Z. For each $\beta \in \mathbb{R}$ we define the following pair of subcategories on $\mathcal{A}$:

$$\mathcal{T}^\beta(A, Z) = \mathcal{T}^\beta = \{\mathcal{E} \in \mathcal{A} : M(\mathcal{G}) > \beta \text{ whenever } \mathcal{E} \twoheadrightarrow \mathcal{G}\}$$
$$\mathcal{F}^\beta(\mathcal{A}, Z) = \mathcal{F}^\beta = \{\mathcal{E} \in \mathcal{A} : M(\mathcal{F}) \leq \beta \text{ for every } 0 \neq \mathcal{F} \subset \mathcal{E}\}.$$

Definition 24.7. The *tilt* $\mathcal{A}^* = \langle \mathcal{F}[1], \mathcal{T} \rangle \subset \mathcal{D}$ of $\mathcal{A}$ with respect to $(\mathcal{T}^\beta, \mathcal{F}^\beta)$ is the smallest extension closed full subcategory of $\mathcal{D}$ containing $\mathcal{T}^\beta$ and $\mathcal{F}^\beta[1]$.

$\mathcal{A}^*$ is the heart of a bounded t-structure on $\mathcal{D}$.

Definition 24.8. Let $\mathcal{E}$ be a torsion free sheaf. We say that $\mathcal{E}$ is *Gieseker semistable* if for every subsheaf $F \subset \mathcal{E}$ with $0 < \mathrm{rk}(F) < \mathrm{rk}(\mathcal{E})$,

$$\frac{P_F(m)}{\mathrm{rk}(F)} \leq \frac{P_\mathcal{E}(m)}{\mathrm{rk}(\mathcal{E})} \qquad \text{for } m \gg 0.$$

It is *Gieseker stable* if we replace $\leq$ by $<$. Here $P_\mathcal{E}(m)$ is the Hilbert polynomial of $\mathcal{E}$.

Definition 24.9. Let $\mathcal{E}$ be a torsion free sheaf. We say that $\mathcal{E}$ is *2-Gieseker semistable* if for every subsheaf $F \subset \mathcal{E}$ with $0 < \mathrm{rk}(F) < \mathrm{rk}(\mathcal{E})$,

$$\frac{P_F(m) - \chi(F)}{\mathrm{rk}(F)} \leq \frac{P_{\mathcal{E}}(m) - \chi(\mathcal{E})}{\mathrm{rk}(\mathcal{E})} \qquad \text{for } m \gg 0.$$

It is *2-Gieseker stable* if we replace $\leq$ by $<$.

The Gieseker chamber is the region where this is the relevant stability. The PT chamber is the neighboring region where the PT condition is the relevant one.

Definition 24.10 (PT stability)**.**

Theorem 24.11. *Assume* $\gcd(\mathrm{ch}_0, \mathrm{ch}_1, \mathrm{ch}_2) = 1$. *Then every PT stable object comes from a stable triple* (B, P, η)*:*

1. B *is a 2-Gieseker semistable sheaf with* $\mathrm{hd}(B) \geq 1$*,*
2. P *is a* 0*-dimensional sheaf,*
3. $\eta \in \mathrm{Hom}_{D^b(X)}(P, B[2]) = \mathrm{Ext}^2(P, B)$ *is nontrivial.*

Moreover,

$$A_\eta := \mathrm{cone}(\eta)[-1].$$

Remark 24.12. Stable triples generalize rank 1 PT stable points.

Theorem 24.13 (Jardim et al.)**.** *There is a single wall crossing that takes the DT moduli space to the PT moduli space. Equivalently, the Gieseker and PT chambers are separated by one wall.*

Proof.

1. Understand the stable objects at the boundary, labelled Θ_v^- in the picture.
2. Study the asymptotic stable objects, that is, what you find if you go to infinity.
3. Identify what the PT chamber is. This may also be an asymptotic analysis.
4. Show that there are no other walls between the Θ wall and infinity on the Gieseker side, and no other walls in the direction of the PT side.

□

History.

- Pandharipande and Thomas (2009) suggested that DT/PT correspondence could be a wall crossing phenomenon in Bridgeland stability space.
- Bayer (2009) established that the rank 1 case is a wall crossing for polynomial stability conditions.
- Toda.
- ...

Case studies.
Let X be a smooth threefold with Picard rank 1.

- Collapsing wall. For $v = (r, 0, 0, -n)$,
$$\mathcal{G}(v) = \mathrm{Quot}^n(\mathcal{O}_X^{\oplus r})//\mathrm{GL}(r),$$
while $\mathcal{T}(v) = \emptyset$.
- Fake wall. For $v = (v_0, v_1, v_2, v_{3,\max})$, $\mathcal{G}(v) = \mathcal{T}(v)$. This is a wall that changes nothing.

Now set $X = \mathbb{P}^3$.

- For $v = (2, -1, -\frac{1}{2}, -\frac{1}{6})$, $\mathcal{G}(v)$ is birational to a $\mathbb{P}^1$-fibration over $\mathbb{P}^3 \times \mathbb{P}^3$, while $\mathcal{T}(v) \cong \mathbb{P}^3$. The wall crossing induces a retraction from the 6-dimensional space $\mathcal{G}(v)$ to $\mathbb{P}^3 \times \mathbb{P}^3$, and $\mathcal{T}(v) \simeq \mathbb{P}^3$.

There is a map

$$\begin{tikzcd} \mathcal{G}(-v) \arrow[dr, "\gamma"] & & \mathcal{T}(v) \arrow[dl, "\tau"'] \\ & \mathcal{M}_{\bar{\beta},\bar{\alpha},s}(v) & \end{tikzcd}$$

Pavel and Tajakka show that the PT moduli stack and the Bridgeland moduli stack at the wall are projective schemes. So the map above is a map of schemes. Formula of the map missing.

Some questions:

- (Existence.) Given a projective variety X, are there stability conditions on $D^{\mathrm{b}}(X)$? Positive answer for surfaces, Fano 3folds with Picard rank 1, abelian 3folds, quintic 3folds, products of curves, $\mathbb{P}^n$, . . .
- (Moduli spaces.) Is $M_\sigma(v)$ a projective scheme? Cannot use usual git techniques to study. Yes for K3, $\mathbb{P}^2$, and $\mathbb{P}^1 \times \mathbb{P}^1$. The parameter space is a stack (ref. missing, involves Toda).
- What do stable objects look like?

Following are very sketchy notes picked up along the way; don't hessitate to skip

(1) Stable objects in an abelian category are the "building blocks": we can reconstruct the whole category from them.
(2) An abelian subcategory (hart) $\subset$ a triangulated
(3) Stability defined via stability f unction on $\mathcal{A}$.
(4) (Example.) $\mathcal{A} = \mathrm{Coh}X$ is heart of $D^b(X)$ with stability function.
(5) Bridgeland Stabl:= the stability conditions are a complex manifold of complex codimension rkΛ:

$$\mathcal{Z} : \mathrm{Stab}(\mathcal{T}) \longrightarrow \mathrm{Hom}(\Lambda, \mathbb{C})$$

(6) There's a chamber structure; moduli space changes across chambers.
(7) Picture: blue + black are walls. Q. What are β and α?
(8) thm: bridgeland stable = gieseker stable ?
(9) Q. slope stability = bridgeland stability? A. Not always.
(10) DT/PT correspondance: only one wall between PT and G chambers
(11) Polynomial stability function. This is an asymptotic version of BS.
(12) There are some ρ's. Arrangements of ρ_i are polynomial stability conditions on a threefold.
(13) Pata-Thomas introduced stability for rank 1 objects. Bayer compares the— wall. Q. Same for Bridge S—only one wall?
(14) Def. A *stable triple*: when $\gcd(\mathrm{ch}_0, \mathrm{ch}_2, \mathrm{ch}_3) = 1$, every PT stable object comes from three conditions (missing).
(15) What happens when you cross the blue wall? Both $\mathcal{G}$ and $\mathcal{T}$ are projective.

(16) For X smooth threefold with rkPic $= 1$, $\mathcal{G}(v)$, $v = (r, 0, 0, -n)$, $\mathcal{G}(v)$ is a known sheaf object and $\mathcal{T} = \emptyset$.
(17) X sm 3 rkpic1, Fake wall; $\mathcal{G} = \mathcal{T}$.
(18) (Extra.) red circle is a wall for a weaker form of stability.

Definition 24.14 (Talk at impa)**.** A rank-2 sheaf $\mathcal{F}$ is *semistable (stable)* if $H^0(\mathcal{F}(t)) = 0$ for $-t \geq (>) \frac{c_1 F}{2}$

Compare with

Definition 24.15 (moduli-curves.tex)**.** Let $f : X \to S$ be a family of curves. We say f is a *semistable family of curves* if

(1) $X \to S$ is a prestable family of curves, and
(2) X_s has genus ≥ 1 and does not have a rational tail for all $s \in S$.

25. Instantons

- The twistor diagram.

$$\begin{tikzcd}
& \ell_\infty \arrow[dl] \arrow[dr, dotted] & \\
\mathbb{C}P^2 \arrow[rr, hook] & & \mathbb{C}P^3 \arrow[dd, "\substack{\text{twistor}\\ \text{map}}"'] \\
& & \\
\mathbb{C}^2\mathbb{R}^4 \arrow[uu, hook] \arrow[rr] & & S^4
\end{tikzcd}$$

- Instantons are solutions to some Yang-Mills solution.
- Expository paper of Donaldson arXiv:2205.08639
- ADHM construction, 1978. The first appearance of Algebraic Geometry in Mathematical Physics. See Hitchin-Kobayashi correspondence.
- Donaldson: "unashamedly computational".
- Expository paper by Simon.
- Take bundle (E, ∇) with an anti-self dual (ADS) connection on S^4 and pullback to $\mathbb{C}P^3$ via the twistor map

$$\tau[x : y : z : w] = [x + jy : z + jw] \qquad \text{note: } \tau^{-1}(p) = \mathbb{C}P^1$$

  Then:
  - Restriction to fibres are trivial.
  - Invariant under anti holomorphic involution (check this formula!) $[x : y : z : w] \mapsto [-y : x : -w : z]$.
  - $\mathcal{E}$ is also an instanton bundle.
  - Penrose Transform: $H^1(\mathcal{E}(-2)) \cong \operatorname{Ker} \Delta = 0$ where Δ is a Laplacian.
  - Definition of instanton sheaf on $\mathbb{P}^3$ via $c_1(E) = 0$ and some vanishing of cohomologies.
  - Passage from [differential equations? algebraic geometry?] to linear algebra: via *monads*. The point is that instanton sheaf is equivalent to "E being the cohomology of a linear monad"; theorem by Horps in the

60's, and is the main tool used by ADHM. Indeed, ADHM equations come from the cohomology sequences of the so-called monads.
– Mathamatical inst bund:= locally free instanton sheaves.

Properties.

- The only instanton of rank 1 on $\mathbb{P}^3$ is the structure sheaf.
- non-trivial rank 2 locally free instanton sheaf is μ_0 stable
- double dual is locally free and also instanton
- non-trivial rank 2 instanton sheaf is Fieseker stable therefore it makes sense to define moduli space of instanton sheaves as an open subset of $\mathcal{M}(c) = \mathcal{G}(k, 0, 2, 0)$.

Then studied the irreducibility (Tikhomirov) and smoothness (Jardim-Verbitsky, 2014. Uses "3rd hyperkähler quotient") of $\mathcal{I}(c)$, the moduli space of rank 2 locally free instanton sheaves of charge c. But nobody likes this results; want new proofs.

In contrast, $\mathcal{M}(c)$ of rank 2-instanton sheaves of charge c is not irreducible in general! $\mathcal{M}(1)$ and $\mathcal{M}(2)$ are irreducible, $\mathcal{M}(3)$ has exactly 2 irreducible components of dimension 21; $\mathcal{M}(4)$ has 4 irredicuble components: the locally free is irreducible, and the other 3 that intersect the closure of the locally free, $\overline{\mathcal{I}(c)}$. 3 components of dimension 29 and one of dimension 32

Remark 25.1. In general the μ moduli space is not projective, but the Gieseker is.

Is $\mathcal{M}(c)$ connected? Use $\mathbb{C}^*$ action. True for $c \leq 4$; every component intersects $\overline{\mathcal{I}(c)}$ in this range!

Definition 25.2 (Elementary transformation)**.** F of rank 2 locally free instanton, Q of rank 0 instanton with 1 dimensional sheaf $h^p Q(-2)$ for $p = 0, 1$. So in this conditions if we have an epimorphism

$$F \overset{\varphi,\,\text{surj}}{\rightarrow}\} Q$$

we get that $\operatorname{Ker}\varphi$ is an instanton.

So we might be interested in

$$E \hookrightarrow E^{**} \xrightarrow{\text{surj.}} Q_E$$

Let's have a look at $\mathcal{M}(3) = \overline{\mathcal{I}(3)} \cup \overline{C(0,1,3}$. Consider a generic line bundle of degree 0 on a planar cubic (which is encoded in that we have intersection of something of cimension 1 and something of simension 3), $Linn \mathrm{Pic}^0(C)$ so we have an epimorphism

$$0 \longrightarrow E \longrightarrow \mathcal{O}_{\mathbb{P}^3} \oplus \mathcal{O}_{\mathbb{P}^3} \longrightarrow (i_* L) \longrightarrow 0$$

yielding a bundle E as previously outlined.

So we are studying the components via pushforwards of sheaves on complete intersection curves inside $\mathbb{P}^3$

"when we do alementary transformation of rational (not rationl?) we get something on the boundary of locally free."

Perverse instanton sheaves. Like an instnaton sheaf in monad description but with some restrictions on the cohomologies. This leads to definition of 0-*rimensional instanton*, a perverse instanton such that $\mathcal{H}^0 = 0$.

Instantons and quivers.

Framed instantons. Fix a line $j : L \to \mathbb{P}^3$ and E a perverse sheaf; a *framing* at L is an isomorphis [missing].

Apply GIT to the ADHM data to construct a moduli space

$$\mathcal{P}(r, c) = \mathcal{V}(r, c)^{\mathrm{st}} // \mathrm{GL}(V_c)$$

and it will follow that $\mathbb{P}(?, ?)$ is connected.

Considerations on quaternionic spaces lead to generalization of what has been discussed so far to higher dimensions. I.e. an *instanton sheaf* on $\mathbb{P}^n$ is... [other cohomological conditions] sheaf

Why are instantons interesting? They are the simplest; may provide examples for Bridgeland stability.

In Kuznetsov (2012) and Faenzi (2014) introduced *rank 2 instanton bundles on Fano 3-folds*. An *instanton bundle* on X is a μ-stable ... and some Chern class is called the *charge*. An *instanton sheaf* (introduced by Marcos-Gaia) is ....

"Since we are imposing μ-stability on the defintion we can consider the moduli $\mathcal{I}_X(c) \subset \mathcal{G}_X(2, -r_X, c, 0)$".

There's also monad representations as an ingredient.

Here are two questions that invite us to join the instanton fever:

Task 1. Construct rank 2-instanton sheaves that do not deform into locally free ones, and obtain the new irreducible components of $\mathcal{G}(2, -r_X, c, 0)$.

Task 2. Nonlocally instanton sheaves that can be deformed into non-locally free ones: the *instanton boundary* $\overline{\mathcal{I}(c)}/\mathcal{I}(X)$[formula right?]

Recipe to construct your own instanton.

(1) (Make a bunch of instantons.) Find an appropriate curve to use Serre correspondence to find some rank 2 instanton sheaf:

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^3}(-1) \longrightarrow \underbrace{E}_{\exists} \longrightarrow \mathcal{I}_{\sqcup \text{lines}} \longrightarrow 0$$

where E is a locally free instanton of charge = the number of lines -1.

(2) (Is your family of instantons generic?) You look at the Exts. "Therefore, the family of instantons only defines a locally closed subset within a generically smooth irreducible component of $\mathcal{G}$".

(3) "Find suitable rank 0 instanton sheaves to perform an elementary transformations on the examples obtained in Step 1." But they are non-locally free.

(4) Now I have my instantons, I know they are locally free: but how to prove that the elementary transformations deform to locally free ones? Looks like the challenge is to prove that the deformation is locally free.

In the papers by the group there are several particular cases when the deformations are locally free. But they don't have a general result that would work for 3-folds.

What you need to call a thing an instanton.

- Minimal cohomology possible; try to kill as much cohomology as you can.
- Fixing c_1 (which may determine other Chern classes).
- Some stability condition like μ-stability or quiver stability. Here is an example that is not μ-semistable: $T\mathbb{P}^3(-1) \oplus \Omega_{\mathbb{P}^3}(1)$

- Whenever possible, look for a monadic representation. (The monadic representation comes from ADHM — the beginnings of this theory. And it's still here!)

26. Jardim-Muniz

Reflexive sheaves are vector bundles except for a small locus.

Definition 26.1. Let X be an integral locally Noetherian scheme. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module. The *reflexive hull* of $\mathcal{F}$ is the $\mathcal{O}_X$-module

$$\mathcal{F}^{**} = \mathcal{H}om_{\mathcal{O}_X}(\mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{O}_X), \mathcal{O}_X)$$

We say $\mathcal{F}$ is *reflexive* if the natural map $j : \mathcal{F} \longrightarrow \mathcal{F}^{**}$ is an isomorphism.

Lemma 26.2. *Let X be an integral locally Noetherian scheme. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module.*

(1) *If $\mathcal{F}$ is reflexive, then $\mathcal{F}$ is torsion free.*
(2) *The map $j : \mathcal{F} \longrightarrow \mathcal{F}^{**}$ is injective if and only if $\mathcal{F}$ is torsion free.*

Remark 26.3 (Talk at IMPA, 11 June). Torsion could also be defined so that the sheaf can inject onto its dual. In this talk we discussed the moduli space of reflexive/torsion-free sheaves, which turned out to be parametrized by c_1, c_2 and c_3. This was denoted by $R(c_1, c_2, c_3)$. Actually I think it may have been Manolache that proved the existence of this moduli space.

Alan Muniz

Nesta palestra discutiremos a classificação de feixes reflexivos de posto dois e seus espaços de módulos. Apresentaremos algumas ferramentas básicas usadas na construção e determinação de tais feixes. Aplicaremos estas técnicas para o caso de feixes com segunda classe de Chern igual a quatro, obtido recentemente em colaboração com Marcos Jardim.

Now let's discuss distributions on manifolds.

Here's the abstract from a talk by Marcos Jardim at Geometric Structures:

"I will revise the work done over the past 10 years with various collaborators on distributions and foliations on 3-folds, especially on the projective space, with a focus on properties of the tangent sheaf and singular scheme."

Here are two key ideas: if the distribution is codimension 1 we can write:

$$0 \longrightarrow F \longrightarrow TX \xrightarrow{\omega} I_Z \otimes L \longrightarrow 0$$

where L is a line bundle and $\omega \in H^0(\Omega_X \otimes L)$, and $Z = \{p : \omega(p) = 0\}$.

When codimension is 2 then $\mathcal{D}$ is given by a holomorphic vector field ν: $T_p = \langle \nu(p) \rangle$.

It can be encoded as an exact sequence

$$0 \longrightarrow L \xrightarrow{\nu} TX \longrightarrow N \longrightarrow 0$$

where L is a line bundle and $\nu \in H^0(TX \otimes L^\vee)$; $Z = \{p | \nu(p) = 0\}$.

Remark 26.4. Saturation means that $Z \subset X$ is a union of curves and points.

And again, distributions are parametrized by Chern classes.

Two interesting open questions:

(1) **Conjeture.** if $\mathcal{D}$ is a codimension 1 foliation of degree d on $\mathbb{P}^3$, then $c_2(F) \leq d^2 - d + 1$ and bound is attained by rational foliations of type $(1, d+1)$. (True for $d \leq 2$.)
(2) **Conjecture (with Pepe Seade).** $\mathcal{D}$ is a codimension 1 foliation on a smooth projective 3-fold, then Sing$\mathcal{D}$ is connected.

Theorem 26.5 (Jardim-Muniz)**.** *Conditions on Chern classes used to understand moduli space $R(c_1, c_2, c_3)$. $c_2 = 4$ gives (?). For $c_3 \leq 6$, possible "spectrum" exists...*

27. Campana

A *special* variety should be thought of as the farthest notion possible from being of general type.

Definition 27.1 (Campana)**.** X complex projective. X is *special* if $\forall i \leq p \leq \dim X$, for all line bundle $L \subset \Omega_X^P$, $\kappa(L) < P$ (Kodaira dimension).

A conjecture by Campana says that X is special if and only if there exists $f : \mathbb{C} \to X$ Zariski dense. Compare this with Kobayashi hyperbolicity! There are no lines on ...?

The main tool for proving this kind of "Bloch-Ochai" statements (i.e. something of the kind: if $h^0(\Omega_X) > \dim X$ then there are no nonconstant $f : \mathbb{C} \to X$ Zariski dense) is by passing through the Albanese variety.

In a talk [fill details] we study a theorem of this kind but with the bound given by $q^+(X)$ defined as the supremum of the $h^0(\hat{X}, \pi^*\Omega_X$ for $\hat{X} \to X$ some (étale) cover. Can we prove a BO stement with the condition $q^+(X) > \dim X$?

Notion of special comes in: indeed

$$q^+(X) > \dim X \implies X \text{ not special } \implies \nexists f : \mathbb{C} \to X \text{Zariski dense.}$$

The following theorem is not at all easy to prove:

Theorem 27.2 (Campana)**.** $\hat{X} \xrightarrow{\text{étale}} X$*, then* $\hat{X}$ *is special iff* X *is special.*

My question: why does the abelianess of the Albanese variety is important in this construction?

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